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**A STRUCTURED APPROACH TO RISK
ANALYSIS IN CONSTRUCTION
PROJECTS USING ISM AND MICMAC
METHODS**

Graduation Project

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ABSTRACT

Construction projects inherently involve a large number of uncertainties and interrelated risk factors, which necessitate the adoption of an effective and systematic risk management approach. However, traditional risk assessment methods generally address risks on an individual basis and fail to adequately consider the structural relationships among them. This limitation significantly constrains decision-making processes, particularly in complex and large-scale projects. In this study, a holistic risk analysis approach that emphasizes the interrelationships among risks in construction projects has been developed.

Within the scope of the study, real project data obtained from an Entegre Project Management company with extensive experience in numerous construction projects across Turkey are utilized. A total of 479 risk records from ten different projects are analyzed, and risks with similar characteristics are classified under 20 potential state categories. To identify cause–effect relationships among risks and to reveal their hierarchical structure, the Interpretive Structural Modeling (ISM) method is employed, while the impact and dependency levels of risks are evaluated using MICMAC (Matrice d'Impacts Croisés Multiplication Appliquée à un Classement) analysis. Furthermore, the classical Risk Priority Number (RPN) method is integrated into this framework, enabling the prioritization of risks not only based on severity, occurrence, and detectability criteria, but also by considering their structural positions within the overall system. Finally, a decision support system named PREVIA (Project Risk Evaluation and Visual Impact Analyzer) is also developed, within the scope of this project using the Python programming language. This system allows risks to be analyzed and visualized together with their hierarchical levels, mutual interactions, and priority rankings.

The findings of the study demonstrate that the integration of the classical RPN approach with the ISM and MICMAC methods enhances risk management in construction projects by making it more structured, consistent, and effective for the decision-makers working on risk response plans.

ÖZET

İnşaat projeleri, çok sayıda belirsizlik ve birbirleriyle etkileşim hâlinde olan risk faktörleri içermeleri nedeniyle etkin ve sistematik bir risk yönetimi yaklaşımını zorunlu kılmaktadır. Ancak geleneksel risk değerlendirme yöntemleri, riskleri çoğunlukla bireysel olarak ele almakta ve riskler arasındaki yapısal ilişkileri yeterince dikkate almamaktadır. Bu durum, özellikle karmaşık ve büyük ölçekli projelerde karar verme süreçlerini sınırlandırmaktadır. Bu çalışmada, inşaat projelerinde riskler arası ilişkileri ön plana alan bütüncül bir risk analiz yaklaşımı geliştirilmiştir.

Çalışma kapsamında, Türkiye genelinde çok sayıda inşaat projesinde görev almış Entegre Proje Yönetimi firmasından temin edilen gerçek proje verileri kullanılmıştır. On farklı projeye ait toplam 479 adet risk kaydı analiz edilerek, benzer özellikler taşıyan riskler 20 adet olası durum başlığı altında sınıflandırılmıştır. Riskler arasındaki neden–sonuç ilişkilerinin belirlenmesi ve hiyerarşik yapının ortaya konulması amacıyla Yorumlayıcı Yapısal Modelleme yöntemi kullanılmış; risklerin etki ve bağımlılık düzeyleri MICMAC analizi ile değerlendirilmiştir. Ayrıca, klasik Risk Öncelik Numarası (RPN) yöntemi bu yapıya entegre edilerek risklerin yalnızca şiddet, olasılık ve fark edilebilirlik kriterlerine göre değil, aynı zamanda sistem içerisindeki yapısal konumları dikkate alınarak önceliklendirilmesi sağlanmıştır.

Geliştirilen entegre metodoloji, Python programlama dili kullanılarak oluşturulan ve PREVIA (Project Risk Evaluation and Visual Impact Analyzer) adı verilen bir karar destek sistemi aracılığıyla uygulanmıştır. Bu sistem sayesinde riskler, hiyerarşik seviyeleri, karşılıklı etkileşimleri ve öncelik sıralamaları ile birlikte analiz edilebilmekte ve görselleştirilebilmektedir. Çalışma sonucunda elde edilen bulgular, klasik RPN yaklaşımının ISM ve MICMAC yöntemleriyle entegre edilmesinin, inşaat projelerinde risk yönetimini daha yapısal, tutarlı ve karar vericiler açısından daha etkin hâle getirdiğini göstermektedir.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	ii
ABSTRACT.....	iii
ÖZET	iv
LIST OF FIGURES	vii
LIST OF TABLES	viii
LIST OF SYMBOLS	ix
1. INTRODUCTION	1
2. LITERATURE REVIEW	3
2.1. Project Management	3
2.2. Risk Management	5
2.3. Releated Studies in Literature	6
3. PROBLEM DEFINITION.....	12
4. METHODOLOGY	14
4.1. ISM – Interpretive Structural Modelling	14
4.1.1. Determination of Risk Factors	15
4.1.2. Determination of Contextual Relationships Among Factors	16
4.1.3. Construction of Structural Self-Interaction Matrix (SSIM).....	16
4.1.4. Conversion of SSIM to Initial Reachability Matrix (T)	16
4.1.5. Transitivity Check and Construction of the Final Reachability Matrix (T*)	17
4.1.6. Calculation of Reachability, Antecedent, and Intersection Sets	18
4.1.7. Level Partitioning and Construction of the ISM Diagram.....	18
4.2. MICMAC – Cross Impact Matrix Multiplication Applied to Classification...	19
4.2.1. Obtaining the Final Reachability Matrix (T*) via the ISM Method.....	20
4.2.2. Using the Final Reachability Matrix in MICMAC	21
4.2.3. Construction of the MICMAC Diagram	21
4.2.4. Classification of Factors into Four Categories	21
4.3. A New Approach for Risk Analysis: Integration of RPN and MICMAC	22
4.3.1. RPN – Risk Priority	22

4.3.2.	An Integrated Approach for RPN and MICMAC Analysis.....	23
5.	IMPLEMENTATION AND RESULTS.....	26
5.1.	Data Collected.....	26
5.2.	Implementation of ISM.....	31
5.2.1.	Establishing Contextual Links Between Factors	31
5.2.2.	Transformation of the SSIM into the Initial Reachability Matrix (T)	32
5.2.3.	Evaluation of Transitive Relationships and Generation of the Final Reachability Matrix (T*)	34
5.2.4.	Calculation of Reachability, Antecedent, and Intersection Sets	38
5.2.5.	Level Partitioning and Hierarchical Diagram	39
5.3.	Implementation of MICMAC	46
5.3.1.	Graphical Representation of Risk Interactions via MICMAC Analysis..	48
5.3.2.	Classification of Risk Factors into Four Categories	49
5.4.	Implementation of a New Approach for Risk Analysis: Integration of RPN & MICMAC	50
5.4.1.	Graphical Representation of Risk Interactions via MICMAC Analysis..	50
5.4.2.	Implementation of a New Approach: RPN and MICMAC Analysis	53
5.4.3.	Development of the PREVIA: Decision Making Support Tool for Risk Management	55
6.	CONCLUSION.....	61
	REFERENCES.....	63
	APPENDIX A- PROJECT SCHEDULE	66
	APPENDIX B - PREVIA SYSTEM SCRIPTS	67

LIST OF FIGURES

Figure 1. Methodology of ISM	15
Figure 2. MICMAC methodology flowchart.....	20
Figure 3. Classification of potential risk.....	26
Figure 4. ISM diagram.....	46
Figure 5. Graph of the MICMAC diagram	48
Figure 6. Workflow of the PREVIA risk-based decision support tool.....	56
Figure 7. PREVIA interface view	57
Figure 8. Table of filtered risk factors	57
Figure 9. RPN section of selected risk factors	58
Figure 10. Visualization of risk analysis and interdependencies in PREVIA	59

LIST OF TABLES

Table 1. Theses in the literature utilizing the ISM-MICMAC method	7
Table 2. Definition and categorization of project risk scenarios	27
Table 3. SSIM matrix.....	32
Table 4. Adjacency matrix (A)	33
Table 5. Initial reachability matrix (T)	34
Table 6. T^2 matrix	35
Table 7. T^3 matrix	36
Table 8. T^4 matrix	37
Table 9. Reachability, antecedent, and intersection sets.....	38
Table 10. First iteration.....	40
Table 11. Second iteration	41
Table 12. Third iteration	42
Table 13. Fourth iteration	43
Table 14. Fifth iteration	44
Table 15. Sixth iteration.....	44
Table 16. Seventh iteration	45
Table 17. Level partitioning.....	45
Table 18. Final reachability matrix DR&DP	47
Table 19. Categories of the risk factors	49
Table 20. Risk priority number (RPN) for the risk factors	52
Table 21. Integrated RPN–MICMAC risk prioritization results (RPN_{DR})	54

LIST OF SYMBOLS

i, j	Indices representing risk factors
n	Total number of risk factors
A	Adjacency matrix
I	Identity matrix
T	Initial reachability matrix
T^*	Final reachability matrix
R_i	Reachability set of risk i
A_i	Antecedent set of risk i
C_i	Intersection set of risk i
DR	Driving power
DE	Dependence power
S	Severity
O	Occurrence
D	Detectability
RPN	Risk Priority Number
RPN_{DR}	Integrated RPN–MICMAC risk priority score
PMI	Project Management Institute
$PMBOK$	Project Management Body of Knowledge

1. INTRODUCTION

Construction projects are dynamic systems characterized by a high level of uncertainty, in which numerous technical and managerial processes are carried out simultaneously (Karadayı, 2014; Gan et al., 2018). Disruptions that arise during the project lifecycle often cannot be attributed to a single cause; instead, they may evolve into complex problems as a result of the interaction and mutual triggering of risks originating from different sources (Shoar & Chileshe, 2021). This situation necessitates addressing risks not merely as isolated events, but as interacting elements within the overall project system.

In current practice, risk assessment activities are predominantly based on individual evaluations of risks in terms of their probability of occurrence and potential impact. While this approach contributes to determining the relative importance of risks, it often falls short in explaining the relational structures underlying schedule delays, cost overruns, or quality-related problems encountered during the project process (Yıldırım, 2022; Çakırlı et al., 2020). Particularly in large-scale projects, the cascading effects caused by the simultaneous activation of multiple risks may exceed the predictive capacity of traditional assessment frameworks (Fang et al., 2012).

The insufficient consideration of interrelationships among risks can lead to the misprioritization of critical risks and the inefficient allocation of limited resources (Karadayı, 2014). Even if a risk appears to be of low significance when considered in isolation, it may assume a decisive role in the project due to its potential to trigger other risks and amplify their impacts across the system (Gan et al., 2018). This highlights the need to move beyond approaches that focus solely on outcomes in risk management, toward perspectives that make cause–effect relationships and structural linkages more explicit.

In response to this need, approaches that position risks within the project system and aim to understand their interactions have gained increasing importance in recent studies (Luthra et al., 2011; Pfohl et al., 2011). Questions such as under what conditions risks emerge, which risks they influence, or by which risks they are affected not only shape the prioritization of risks but also directly influence strategic decision-making in project management (Yomralıoğlu, 2022). In this context, risk management should be regarded not merely as an

evaluation tool, but as an analytical framework that actively guides decision-making processes.

This study begins with the identification of risks encountered in construction projects and aims to address these risks in more meaningful clusters based on shared characteristics. Subsequently, cause–effect relationships and mutual interactions among risks are examined, and the structural implications of these relationships within the project system are revealed. The resulting relational structure enables a clearer understanding of the roles of risks within the system and provides a foundation for deepening the risk evaluation process. In the final stage, considering risks in light of their structural positions is intended to bring a more holistic and multi-layered perspective to risk management.

This study consists of interrelated sections that collectively contribute to the systematic development and implementation of a holistic risk management framework for construction projects. Following the introduction, the second section presents a comprehensive literature review that examines traditional risk assessment approaches used in construction projects and focuses on contemporary methods that address interdependencies among risks. The third section describes the data set and problem context by detailing the characteristics of the construction projects analyzed and the classification of identified risks. The fourth section outlines the proposed methodology, explaining the integration of the classical Risk Priority Number (RPN) approach with Interpretive Structural Modeling (ISM) and MICMAC (*Matrice d'Impacts Croisés Multiplication Appliquée à un Classement*) analysis, as well as the decision support system (PREVIA) developed using the Python programming language. In the subsequent section, the application of the proposed framework is presented, and the analysis results are discussed in terms of risk hierarchy, mutual interactions, and prioritization. Finally, the conclusion summarizes the main findings of the study, highlights the contributions of the integrated approach to decision-making processes in construction risk management, and offers recommendations for future research. The appendix presents supporting materials, including the project schedule obtained from MS Project and selected PREVIA system scripts, which complement the methodological and implementation details discussed in the main text.

2. LITERATURE REVIEW

2.1. Project Management

Project management is the process of planning, executing, and controlling temporary endeavors undertaken to create a unique product, service, or result, subject to specific constraints of time, cost, and scope (PMI, 2021). Projects are a crucial vehicle for organizations to achieve their strategic objectives, and an effective project management approach facilitates the efficient utilization of resources, mitigation of risks, and assurance of stakeholder satisfaction (Kerzner, 2017).

The Project Management Body of Knowledge (PMBOK) Guide, published by the Project Management Institute (PMI), establishes the global standards for the discipline of project management. First published in 1987, the Guide, now in its seventh edition, has introduced a principle-based perspective to project management (PMI, 2021). The PMBOK organizes project management around five process groups (Initiating, Planning, Executing, Monitoring and Controlling, and Closing) and ten Knowledge Areas.

These knowledge areas are integrated throughout the project lifecycle to ensure a balance among core objectives such as scope, schedule, cost, quality, and risk. Each knowledge area is defined by specific processes, inputs, tools, techniques, and outputs (Jainendrakumar, 2015).

The ten knowledge areas of project management are elaborated below:

Project Integration Management

This is the knowledge area aimed at achieving project objectives by ensuring all processes and components, from project charter development to execution and change control, are planned coherently and managed holistically (PMI, 2021; Hornstein, 2015).

Project Scope Management

This area ensures that only the necessary work is performed by defining the objectives, deliverables, and the work required. Through processes such as scope planning, requirements collection, creating the Work Breakdown Structure (WBS), and scope control, it aims to prevent uncontrolled expansion known as "scope creep" (Tuuli et al., 2023; Pheng, 2018).

Project Schedule Management

This knowledge area ensures the timely completion of the project by defining, sequencing, and planning the durations of activities. Techniques such as Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT) are utilized in this process to mitigate delays and enhance delivery performance (PMI, 2017; Heagney, 2016).

Project Cost Management

This area aims to ensure the project's profitability and sustainability through the planning, estimating, budgeting, and controlling of financial resources, often employing Earned Value Management (EVM) techniques (Hansen, Mowen & Heitger, 2021).

Project Quality Management

This area encompasses planning, quality assurance, and control processes to ensure that products and processes comply with defined standards. Approaches like Total Quality Management (TQM) and Six Sigma are applied to guarantee quality, thereby reducing the cost of rework and increasing stakeholder confidence (Berényi, 2023; Wolniak, 2022).

Project Resource Management

This area aims to optimize team performance, clarify role assignments, and prevent resource conflicts by planning, allocating, and optimizing human, material, and equipment resources. Techniques like Resource Leveling and Resource Smoothing are used to create a balanced workload plan (Heagney, 2016).

Project Communication Management

Communication management plays a critical role in ensuring the correct, timely, and effective flow of information within projects. Communication planning, information distribution, and performance reporting are the core components of this area (Zulch, 2014).

Project Risk Management

Risk management involves the identification, analysis, and control of uncertainties that may arise during the project lifecycle. The processes include risk planning, identification, qualitative/quantitative analysis, response planning, and risk monitoring (PMI, 2017).

Project Procurement Management

Procurement management plans and oversees the acquisition of goods and services required for the project from external sources. Purchasing processes, contract administration, and monitoring of supplier performance are included in this knowledge area (Thammasi, 2018).

Project Stakeholder Management

Stakeholder management involves managing the needs and expectations of all parties who are affected by or can affect the project. This process includes identifying stakeholders, developing engagement plans, and monitoring communication and participation levels (Eskerod, Huemann & Savage, 2015). It helps garner project support and minimizes potential conflicts.

The holistic integration of project management knowledge areas ensures the achievement of effective and sustainable outcomes that are aligned with organizational goals; specifically, maintaining the balance among risk, cost, and schedule management is a key determinant of project success (Kerzner, 2017; PMI, 2021).

2.2. Risk Management

The presence of uncertainty, complexity, and dynamic conditions in projects makes the emergence of risks inevitable. Consequently, risk management stands as a critically important function within the project management knowledge areas (PMI, 2021).

Risk is defined as an uncertain event or condition that, if it occurs, will have a positive or negative effect on the project's objectives (Hillson & Simon, 2020). Project risk management is the systematic process of identifying, analyzing, and bringing these uncertainties under control (PMI, 2017). Since construction projects inherently carry numerous technical, financial, environmental, and organizational risks, accurately understanding the interactions among these risks is critical for project success (Zou, Zhang & Wang, 2007; Aven & Renn, 2015). Indeed, effective risk management practices enhance project performance by reducing time and cost variances (Carvalho & Rabechini Junior, 2015).

According to the PMBOK Guide (PMI, 2021), project risk management consists of seven fundamental processes. The project risk management process involves the steps of planning risk management, identifying risks, performing qualitative and quantitative analyses,

developing and implementing appropriate response strategies, and continuously monitoring and controlling risks (PMI, 2017). The systematic execution of this process ensures the early detection of risks and the mitigation of negative impacts (Kerzner, 2017).

Risk management approaches are broadly categorized into qualitative and quantitative methods. While traditional methods offer subjective assessments based on expert opinions, they often fail to adequately reflect the relationships between risks in complex projects (Perry & Hayes, 1985). Therefore, in recent years, structural models that analyze the causal relationships of risks, such as the ISM (Interpretive Structural Modeling) and MICMAC (Matrice d'Impacts Croisés Multiplication Appliquée à un Classement) methods, have been preferred. These approaches classify risks according to their driving power and dependence, enabling the determination of the most critical risks (Sage, 1977; Warfield, 1974).

The construction industry is one of the areas with the highest incidence of risks due to high capital requirements, multi-stakeholder structures, and environmental uncertainties; these risks often culminate in cost overruns, delays, and quality issues (Zou et al., 2007; Ozorhon, 2019). The literature has seen an increase in studies focusing not only on risk identification but also on the analysis of their interrelationships. For instance, Dey and Ogunlana (2004) systematically compared different risk management tools and examined their impact on project success, while Rostamzadeh et al. (2020) utilized MICMAC analysis to classify risks in infrastructure projects based on their driving and dependence power. Such structural analyses provide strategic decision support to project managers by enabling the prioritization of high-impact risks and the efficient utilization of limited resources (Goswami et al., 2019).

2.3. Related Studies in Literature

Interpretive Structural Modeling (ISM) and MICMAC (Cross-Impact Matrix Multiplication Applied to Classification) methods are structured analysis approaches developed to establish the cause-and-effect relationships among variables in complex systems. ISM was developed by Warfield in 1974 to hierarchically model the relationships among factors in social, economic, and technical systems; whereas MICMAC was proposed by Godet in 1986 to evaluate the structural dynamics of the system by analyzing the "influence" and "dependence" levels of the factors. Over the past two decades, these two methods have been adapted to different disciplines, and reviewed studies show that ISM and MICMAC have been enhanced through classical, weighted, or integrated versions applied in various sectors.

In the literature, it is observed that the ISM and MICMAC methods are increasingly utilized for the systematic analysis of risks in complex systems and for revealing the interactions among them. These methods are effectively applied in various sectors to determine cause-and-effect relationships between factors, establish hierarchical structures, and classify the roles of variables within the system.

Table 1 summarizes 15 theses and references from the last 11 years examined in this study, providing a general framework that outlines in which fields, for what purposes, and how the ISM and MICMAC methods have been applied.

Table 1. Theses in the literature utilizing the ISM-MICMAC method

Reference	Application Area	Methods	Scope
Karadayı (2014)	Service Sector	ISM + MICMAC	Risk Management
Çakırlı (2017)	Corporate Processes	ISM + MICMAC	Process Management
Ünlü (2017)	Logistics	ISM + MICMAC	Process Management
Almasri (2019)	Supply Chain	ISM + MICMAC	Risk + Process Management
Yayla (2019)	IT Project Management	ISM	Project Performance Analysis
Guan et al. (2020)	Green Building Projects	ISM + MICMAC	Project Risk Analysis / Sustainability
Çelebi (2021)	Gastronomy / Social Entrepreneurship	ISM + MICMAC	Process Management (Social Innovation)
Mansuroğlu (2021)	Transportation	ISM	Strategic Management
Yomralıoğlu (2022)	Aviation	Weighted ISM + MICMAC	Process / Quality Management
Nikjow et al. (2022)	Infrastructure Projects (Belt and Road)	ISM + MICMAC	Project Risk Analysis / Infrastructure Projects
Sağ (2022)	Open Innovation	ISM	Conceptual Model / Index Development
Yıldırım (2022)	SME / Green Innovation	Fuzzy ISM + MICMAC + DEMATEL	Process Management / Barrier Analysis
Tuncer (2024)	Project Management	ISM + MICMAC + AI	Project + Risk Management
Koçak (2024)	Mobile Payment Systems	ISM + Delphi	Digital Transformation Barriers
Ergün (2025)	Automotive / Production	SMART ISM + MICMAC	Project + Process Management

In the study conducted by Karadayı (2014), a methodological framework using ISM-MICMAC analysis was presented for risk analysis in service supply chains (SSCs). The SSC

risks (finance, IT, macro-level, etc.), identified through expert opinions, were hierarchically modeled with ISM, and their influence/dependence powers were determined with MICMAC. The study positioned macro-level risks as the most independent driving force and information technology risk as a second-level influential variable. This analysis is a significant methodological contribution that systematically demonstrates which risk factors should be prioritized in SSC risk management (Karadayı, 2014).

Two studies conducted in 2017 provide significant examples of the use of ISM-MICMAC methods in the context of process analysis. Çakırlı's (2017) research hierarchically modeled the barriers encountered during an ERP implementation process in Turkey, revealing critical factors such as resistance to change, lack of top management support, and corporate culture incompatibility. Similarly, Ünlü's (2017) study, which addressed transportation mode selection criteria in the logistics sector, utilized the ISM-MICMAC approach to show that the number of customer complaints, transportation cost, product capacity, and service frequency were decisive in the decision-making process. Both studies offer a common methodological contribution to the literature by systematically examining the structural relationships of elements affecting processes in different sectors.

Almasri's (2019) study examined the hierarchy of Critical Success Factors (CSFs) for Sustainable Supply Chain Management (SSCM) and their performance impacts in the Turkish manufacturing industry. In the first stage, Interpretive Structural Modeling (ISM) and MICMAC analysis were used to analyze the relationships among the CSFs. The ISM/MICMAC results identified competition/stakeholder pressure and regulations as external factors with the highest driving power. In the second stage, Structural Equation Modeling (SEM) was applied using survey data from 450 firms; it was confirmed that CSFs have significant positive impacts on SSCM practices and, indirectly, on sustainability performance (social, economic, environmental) (Almasri, 2019).

Yayla's (2019) study analyzed the relationships among the factors affecting the success of IT projects using the ISM method and revealed that organizational structure, organizational culture, and management support are the strongest driving factors. The findings indicated that IT project failures mostly stem from root causes of an organizational nature (Yayla, 2019).

Concurrently, the research by Guan, Abbasi, and Ryan on green building projects used ISM and MICMAC analyses to assess risk dependencies throughout the project lifecycle and

demonstrate how critical risk factors are identified in sustainability-focused projects (Guan et al., 2020). This study demonstrates the effectiveness of ISM-based methods in analyzing complex project risks in a structured manner.

Çelebi's (2021) doctoral thesis utilized ISM and MICMAC methods to evaluate the structural relationships among themes derived from qualitative content analysis, in order to examine the social entrepreneurship phenomenon in the gastronomy sector. The findings revealed significant interactions among social entrepreneurs' personality traits, motivations, the challenges they encountered, and their social contributions. The study is significant in the literature as it demonstrates that the ISM-MICMAC approach can be utilized in the field of social entrepreneurship to develop qualitative data-based conceptual models (Çelebi, 2021).

Mansuroğlu (2021) examined the relationship between the competitive strategies and firm performance of Ro-Ro companies operating in Turkey. The study employed Interpretive Structural Modeling (ISM) and MICMAC analysis to model the relationships between strategy components and performance criteria, which were identified through a literature review and semi-structured interviews. This methodological approach revealed the hierarchical and reciprocal interactions among variables specific to the sector's dynamics. The findings indicated that customer focus, operational flexibility, innovativeness, and R&D are the most critical strategy components, suggesting that managers should adopt hybrid strategies (Mansuroğlu, 2021).

Yomralıoğlu's (2022) study analyzed the interactions among service quality indicators in airline operations using W-ISM (Weighted ISM) and MICMAC methods, identifying the quality components with the highest driving power in the system. Similarly, Nikjow et al. (2022) examined the hierarchical structure and interdependencies of risk factors in large-scale infrastructure projects using the ISM-MICMAC approach, revealing the guiding effects of critical risk elements on the system. Both studies demonstrate that the ISM-MICMAC methods provide a powerful analytical framework for evaluating interaction networks in complex systems.

Sağ's (2022) doctoral thesis modeled the relationships among the identified "erosion factors" using the ISM method to evaluate the open innovation approach, which aims to enhance firms' innovation processes through knowledge sharing and collaboration with external stakeholders. The analysis results indicated that digitalization and globalization are the strongest driving factors for open innovation. The study stands out in the literature for

developing an Open Innovation Index based on the structural model obtained, demonstrating that the ISM method can also be effectively used in conceptual model creation processes (Sağ, 2022).

Yıldırım's (2022) study utilized Fuzzy ISM, MICMAC, and DEMATEL methods together to analyze the barriers faced by SMEs in the process of adopting green innovation. The DEMATEL method, used in the study, is a decision analysis technique that quantitatively evaluates the mutual influence levels and cause-and-effect relationships between factors. The findings indicated that financial inadequacy, lack of management support, and weak technological infrastructure are the most critical barriers in the system. The study makes a significant contribution to the literature by integrating fuzzy logic and DEMATEL into the ISM-MICMAC structure, enabling a more comprehensive evaluation of green innovation barriers (Yıldırım, 2022).

In the study conducted by Tuncer (2024), an AI-supported risk analysis model was developed in the field of project management, with the methodological framework built upon an ISM-MICMAC-AI integration. In the study, risks identified through expert opinions and a literature review were hierarchically modeled using the ISM method; subsequently, the influence and dependence powers of the variables were evaluated with MICMAC analysis. Furthermore, the automation of the process in the MATLAB environment added an innovative dimension to the method. Tuncer's study offers a significant contribution to the literature by demonstrating the applicability of an AI-based ISM-MICMAC approach to project risk management (Tuncer, 2024).

In Koçak's (2024) study, the factors negatively affecting the adoption of mobile payment systems in Turkey were analyzed. More than 50 barriers identified in the literature were narrowed down to 21 fundamental barriers following a Delphi study conducted with experts. The Delphi method was used as a structured survey technique aiming to achieve a controlled consensus by obtaining feedback from experts over multiple rounds. The 21 resulting barriers were hierarchically modeled using Interpretive Structural Modeling (ISM), and age and inadequate technological infrastructure were identified as the strongest driving barriers to the adoption of mobile payment systems (Koçak, 2024).

Ergün (2025) analyzed the enablers for implementing the Lean and Green Management (LG) system for SMEs in the Turkish automotive sector. The study introduced a methodological innovation by integrating SMART ISM software, which eliminates ISM matrix calculation

errors, and MICMAC analysis. The analyses identified supply chain management, green economy initiatives, and energy efficiency as the independent variables (fundamental starting points) with the highest driving power. These results present a systematic hierarchy indicating which fundamental variables SMEs, intending to start their LG transformation, should focus on for strategic resource allocation (Ergün, 2025).

The reviewed theses clearly demonstrate that the ISM and MICMAC methods are reliable and flexible tools for analyzing complex systems across various sectors. In all studies, causal relationships among factors were consistently identified, and hierarchical structures were established to explain the functioning of the systems. While the focal points of the studies varied according to the sector, similar methodological steps were followed in areas such as organizational processes (e.g., ERP), service quality, innovation management, sustainability, and social entrepreneurship. The acquisition of data predominantly through expert opinions and literature reviews indicates that these methods can be applied in conjunction with both qualitative and quantitative approaches. In a significant portion of the studies, MICMAC analysis was also utilized to evaluate the factors' influence (driving power) and dependence (dependence power) levels. These common trends highlight the interdisciplinary applicability of the ISM and MICMAC methods and confirm that they provide a common structural analysis framework across different research areas.

The original contribution of this study is its fulfillment of the gap in the limited literature that holistically examines the interrelationships among risk factors in construction projects. Unlike traditional approaches, this study proposes a data-driven decision support model by integrating ISM, MICMAC, and RPN (Risk Priority Number) methods into a single systematic structure. This integrated methodological framework establishes a more consistent and objective structure for risk management in construction projects.

3. PROBLEM DEFINITION

Effective risk management in project management plays a critical role in ensuring that projects are completed within the planned schedule, budget, and quality objectives. Construction projects, by their very nature, involve numerous uncertainties and risk factors simultaneously; therefore, it is of great importance that these risks are systematically analyzed and managed. However, traditional risk assessment approaches are mostly based on subjective evaluations, which often fail to consider the interrelationships among risks, leading to deficiencies in the decision-making process.

This project has been conducted in collaboration with Entegre Project Management, a company that has participated in numerous large-scale construction and investment projects across Turkey. Established in 2003, the company has successfully managed more than 200 projects with a total construction area exceeding 10 million square meters and has accumulated extensive industry experience through its team of over 160 engineers and architects. With an average of 30 years of experience in project and construction management, the management team possesses in-depth field knowledge regarding risk management processes. Entegre Project Management holds ISO 9001, ISO 14001, and ISO 45001 certifications, providing its services within the framework of internationally recognized systems and procedures. Its areas of expertise include the management of design, tendering, implementation, and delivery processes, as well as cost, contract, and planning consultancy. This experience and organizational structure have provided a strong foundation in terms of the reliability and representativeness of the risk data to be used in this study. Accordingly, the analysis of risks encountered in construction projects has been carried out using data provided by the company.

The aim of the study is to encounter the key risk factors in construction projects, and to analyze the relationships among these risks through Interpretive Structural Modeling (ISM) and MICMAC (Matrice d'Impacts Croisés Multiplication Appliquée à un Classement) methods. Through the ISM method, cause–effect relationships among risks can be identified, and based on these relationships, the hierarchical structure of the risks are established. Subsequently, the MICMAC analysis is applied to evaluate the driving and dependence power of the risks, allowing them to be classified according to their systemic roles. Furthermore, the Risk Priority Number (RPN) analysis is integrated into this framework to

enable the evaluation of risks not only in terms of their structural relationships but also according to their quantitative priorities.

Within the scope of this study, real risk data is used from the records of the company and also an expert team help is provided for the required support. The experts' support is required for the finalization of potential risk factors identified among the company data, assessment of them for the interactions among them, and determination of influences. The resulting relational structure is the primary input for the ISM and MICMAC analyses. Moreover, current risk analysis approach of the company is also integrated to the proposed approach and a decision support software for the ease of analysis is developed using the Python programming language. Thus, ISM, MICMAC, and RPN methods have been integrated within a software environment, enabling the systematic analysis, evaluation, and monitoring of risks.

The primary aim of this project is to develop a more holistic and data-driven decision support system that emphasizes interrelationships among risks in construction projects. This system is designed to make the risk management process more structured, consistent, and reliable, thereby providing companies with a new perspective in analyzing, prioritizing, and focusing on critical risks.

4. METHODOLOGY

To achieve a structured and comprehensive risk system analysis, this study adopts an integrated methodological framework combining Interpretive Structural Modeling (ISM), MICMAC (Matrice d'Impacts Croisés Multiplication Appliquée à un Classement) analysis, and an enhanced Risk Priority Number (RPN) approach. ISM is employed to identify and model the hierarchical interrelationships among risks based on expert judgments, enabling the explicit representation of both direct and indirect interactions within the system. Building upon this structural foundation, MICMAC analysis is used to quantitatively assess these relationships by examining the driving and dependence characteristics of each risk, thereby revealing the underlying influence mechanisms and critical structural roles within the risk network. While ISM and MICMAC focus on relational and structural properties, they do not explicitly account for project-specific risk attributes such as probability, impact, and detectability. To address this limitation, RPN is integrated into the MICMAC-based structure as a complementary component, enriching the analysis by incorporating intrinsic risk characteristics without altering the established interaction framework. Through this integrated approach, the methodology simultaneously captures the systemic influence of risks and their context-dependent criticality, enabling a more realistic, comparable, and decision-oriented prioritization of risks.

4.1. ISM – Interpretive Structural Modelling

Interpretive Structural Modeling (ISM) is a modeling approach that systematically analyzes the relationships among the elements constituting a system and organizes these elements into a hierarchical structure. Developed by Warfield (1974), the method aims to transform systems that initially contain ambiguity or are difficult to define into a more understandable format. ISM facilitates a holistic examination of the system by making the direct and indirect interactions between the factors influencing a problem visible (Sage, 1977).

The core logic of the method is based on the initial identification of relevant factors, followed by the structuring of interrelationships via expert judgments. During this procedure, contextual relationships are concretized incrementally, creating a hierarchical model that indicates the specific level of each factor. This simplification allows for a clear visual and

logical understanding of the system's operation (Mandal & Deshmukh, 1994; Singh et al., 2003).

Since ISM evaluates relationships qualitatively, the model is entirely dependent on expert insights. Therefore, the developed structure is contingent upon the knowledge and interpretation of the experts involved. The method reveals the presence and direction of interactions but does not include a measurement regarding their magnitude or weight. These features demonstrate that ISM allows for a structured and consistent exposition of relationships in complex systems, yet remains sensitive to the validity of expert judgments. The steps of the ISM methodology are presented in Figure 1.

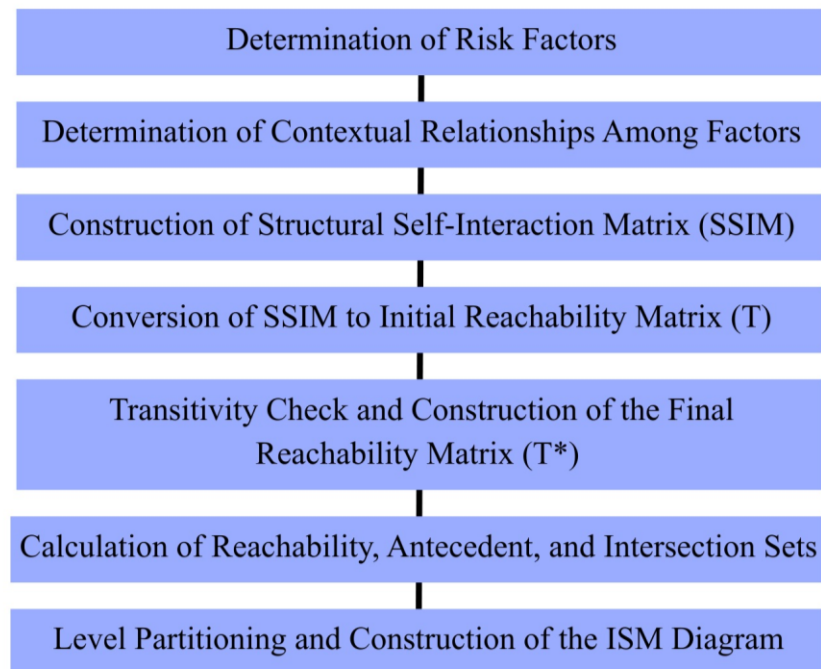


Figure 1. Methodology of ISM

4.1.1. Determination of Risk Factors

All factors affecting the system are defined, and the list of factors to be utilized in the model is generated. At this stage, all factors relevant to the problem area are identified and aggregated into a set:

$$E = \{e_1, e_2, \dots, e_n\}$$

This step is strictly a definition process; it does not involve any mathematical computations.

4.1.2. Determination of Contextual Relationships Among Factors

The relationship between each pair of factors is established through expert judgments. For every pair (e_i, e_j) , the question "Does element i influence element j ?" is evaluated. This contextual relationship is functionally defined as follows:

$$R: (e_i, e_j) \rightarrow \text{relationship type}$$

The existence and direction of the relationship are utilized in the Structural Self-Interaction Matrix (SSIM) in the subsequent step.

4.1.3. Construction of Structural Self-Interaction Matrix (SSIM)

The SSIM is constructed by denoting the relationships between factor pairs using the symbols V, A, X, and O. For each pair of factors, the following symbols are utilized:

V: There is a relationship from e_i to e_j . (one-way only).

A: There is a relationship from e_i to e_j . (one-way only).

X: There is a mutual relationship between e_i and e_j . (bi-directional).

O: There is no relationship between e_i and e_j .

Mathematically expressed as:

$$s_{ij} \in \{V, A, X, O\}$$

These symbols are recorded within the matrix structure $S = [s_{ij}]$.

4.1.4. Conversion of SSIM to Initial Reachability Matrix (T)

The symbols in the SSIM are transformed into binary values (0 and 1) to generate the relationship matrix. The conversion rules are defined as follows:

$$a_{ij} = \begin{cases} 1, & \text{if } s_{ij} = V \text{ or } X \\ 0, & \text{if } s_{ij} = A \text{ or } O \end{cases}$$

$$a_{ji} = \begin{cases} 1, & \text{if } s_{ij} = A \text{ or } X \\ 0, & \text{if } s_{ij} = V \text{ or } O \end{cases}$$

As a result of this operation, the adjacency matrix is obtained:

$$A = [a_{ij}]$$

The relationship between risks themselves is not defined within in the adjacency matrix.

First, an initial transformation is performed by adding the identity matrix to the adjacency matrix. As presented in Equation (1), this operation yields the matrix

$$T = (A + I) \quad (1)$$

which ensures that each factor is assumed to be reachable from itself and provides the basis for incorporating self-influence into the system.

4.1.5. Transitivity Check and Construction of the Final Reachability Matrix (T^*)

The initial matrix is processed in accordance with the transitivity principle—the fundamental assumption of ISM—to obtain the final reachability matrix. Transitivity is essential in ISM:

If $e_i \rightarrow e_j$ and $e_j \rightarrow e_k$, then $e_i \rightarrow e_k$ must hold.

Therefore, the initial reachability matrix is processed using Boolean arithmetic to derive the final reachability matrix.

Boolean addition rules:

$$1 + 1 = 1$$

$$1 + 0 = 1$$

$$0 + 0 = 0$$

Subsequently, the matrix T is subjected to successive Boolean multiplications (powering) in order to account for both direct and indirect relationships among the factors. Through this iterative process, higher-order reachability information is progressively revealed.

As stated in Equation (2), the iteration continues until a stable state is reached, that is, when the condition

$$T^* = T^k = T^{k+1}, k > 1 \quad (2)$$

holds.

Once no further changes are observed between two successive matrix powers, the iterative process is terminated, and the resulting matrix T^* is obtained. This stabilized matrix represents the final reachability matrix and serves as a key input for the subsequent stages of the ISM methodology.

Boolean multiplication:

$$1 \times 1 = 1$$

$$1 \times 0 = 0$$

$$0 \times 0 = 0$$

4.1.6. Calculation of Reachability, Antecedent, and Intersection Sets

Following the derivation of the final reachability matrix, the relative position of each risk within the system is examined through the calculation of reachability, antecedent, and intersection sets. These sets provide a systematic basis for identifying the hierarchical structure of the system and determining the level at which each risk is located.

For a given risk i , the reachability set, denoted as $R(i)$, consists of all factors j that can be reached from i according to the final reachability matrix. In other words, $R(i)$ includes not only the risks that are directly influenced by i , but also those that are indirectly influenced through transitive relationships. This relationship is defined in (3).

$$R(i) = \{j \mid t_{ij} = 1\} \quad (3)$$

Conversely, the antecedent set, denoted as $A(i)$, comprises all risks j that have an influence on factor i . This set captures both direct and indirect influences exerted on i by other risks within the system, as expressed in (4).

$$A(i) = \{j \mid t_{ji} = 1\} \quad (4)$$

The intersection set, $C(i)$, is obtained by identifying the common elements of the reachability and antecedent sets. This set represents the risks that both influence i and are influenced by i , indicating mutual or feedback-based relationships. The intersection relationship is given in (5).

$$C(i) = R(i) \cap A(i) \quad (5)$$

Together, these sets enable a structured and consistent identification of the hierarchical roles of risks within the system, forming the analytical foundation for the subsequent level partitioning process.

4.1.7. Level Partitioning and Construction of the ISM Diagram

These sets constitute the foundation of the level partitioning procedure in ISM. A risk is assigned to the highest hierarchical level when its reachability set is identical to its

intersection set. This condition, expressed in (6), indicates that the risk does not lead to any other risks beyond those with which it shares mutual influence.

$$C(i) = R(i) \quad (6)$$

This condition indicates that the risk does not lead to any other risks beyond those with which it shares mutual influence and, therefore, does not function as a driving element within the system. Such risks are interpreted as outcome-oriented or dependent, as they primarily reflect the effects of other risks rather than acting as sources of influence themselves. Accordingly, risks satisfying this condition are assigned to the corresponding hierarchical level.

Once identified, these risks are fixed at their respective level and subsequently removed from the final reachability matrix to avoid repeated assignment in later stages. The same analytical procedure is then iteratively applied to the reduced matrix to determine the remaining hierarchical levels. Through this systematic process of elimination and reassessment, all risks are progressively allocated to appropriate levels, resulting in a clear and well-defined hierarchical structure of the system.

Following the completion of the level partitioning process, the derived hierarchical structure is translated into a directed ISM diagram. In this diagram, risks are positioned according to their finalized hierarchical levels, with Level 1 placed at the top and successive lower levels arranged beneath it, thereby visually representing the direction of influence within the system.

To enhance clarity and ensure that the diagram reflects only meaningful relationships, transitive links are subsequently eliminated. Although indirect relationships are inherently captured through the reachability matrix and the hierarchical structure, the ISM approach emphasizes the visualization of direct influences. Therefore, links that arise solely due to transitivity are excluded from the final diagram. As a result, the constructed ISM model provides a concise and interpretable visual representation of both the hierarchical ordering of risks and the significant direct interactions among them.

4.2. MICMAC – Cross Impact Matrix Multiplication Applied to Classification

MICMAC (Cross-Impact Matrix Multiplication Applied to Classification) analysis is a structural technique used to evaluate how factors within a system influence each other and

how strongly they depend on other factors. It is particularly useful for analyzing complex, multi-dimensional systems in which interactions are both direct and indirect, making it possible to identify critical drivers, outcome factors, and potential instability zones within the system. MICMAC determines the structural importance of each factor based on two quantitative indicators: driving power and dependence power, which reveal the root causes that shape the system structure as well as those factors positioned as final results.

MICMAC is the natural continuation of the Interpretive Structural Modeling (ISM) methodology. While ISM establishes the hierarchical structure among factors based on expert judgments and the final reachability matrix, MICMAC quantifies the influence mechanisms underlying this hierarchy and highlights feedback loops that may exist. Together, the two methods allow both the structure and the dynamics of the system to be analyzed comprehensively, enabling strategic prioritization and more targeted intervention strategies. The steps of the MICMAC methodology are presented in Figure 2.

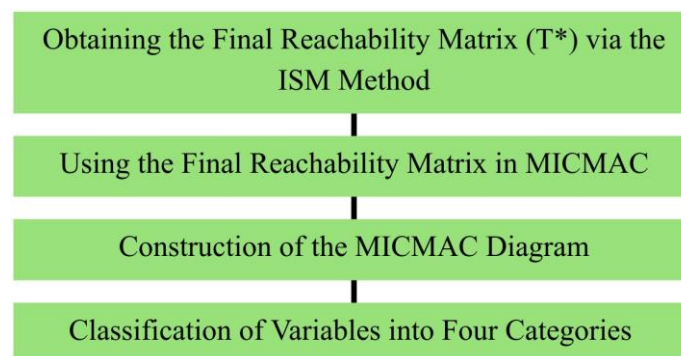


Figure 2. MICMAC methodology flowchart

4.2.1. Obtaining the Final Reachability Matrix (T*) via the ISM Method

The final reachability matrix (FRM) obtained at the end of the ISM process forms the primary input for MICMAC. This matrix incorporates all direct and indirect relationships among the factors. MICMAC processes this matrix mathematically to determine: The degree to which each factor influences other factors (driving power), and the degree to which each factor is influenced by other factors (dependence power). ISM answers “Which factor influences which?” while MICMAC answers “Which factors are the key drivers, which are outcomes, and which are unstable linkages?”

4.2.2. Using the Final Reachability Matrix in MICMAC

The final reachability matrix obtained from the ISM analysis is used as the input for the MICMAC procedure and is represented in (7), where each element fr_{ij} denotes the influence relationship between factors i and j .

$$\text{FRM} = \begin{bmatrix} fr_{11} & fr_{12} & \cdots & fr_{1n} \\ fr_{21} & fr_{22} & \cdots & fr_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ fr_{n1} & fr_{n2} & \cdots & fr_{nn} \end{bmatrix} \quad (7)$$

Based on this matrix, MICMAC computes two measures for each factor. The driving power is calculated as the sum of the elements in the corresponding row, as given in (8), while the dependence power is obtained by summing the elements in the corresponding column, as defined in (9).

Driving Power (DR):

$$DR_i = \sum_{j=1}^n fr_{ij} \quad (8)$$

Dependence Power (DE):

$$DE_j = \sum_{i=1}^n fr_{ij} \quad (9)$$

Driving power is the row sum, and dependence power is the column sum. Together, these two indicators reveal the structural role of each factor. These measures form the basis for the subsequent classification of factors within the MICMAC analysis.

4.2.3. Construction of the MICMAC Diagram

Driving and dependence values are plotted on a two-dimensional diagram.

X-axis: Dependence Power (DE)

Y-axis: Driving Power (DR)

Each factor is placed on the coordinate plane according to its DP and DE values. The location of each point indicates the structural behavior of that factor within the system.

4.2.4. Classification of Factors into Four Categories

Based on their positions in the diagram, factors fall into one of the following categories.

Autonomous Factors (Low DR, Low DE): These factors are weakly connected to the system. They have minimal influence and minimal dependence. Their strategic importance is generally low.

Dependent Factors (Low DR, High DE): These factors depend heavily on others. They represent system outcomes rather than driving mechanisms.

Linkage Factors (High DR, High DE): These factors are both highly influential and highly dependent. They are unstable and sensitive because any change in them affects many others and they themselves are affected by many factors. They require careful monitoring.

Independent/Driving Factors (High DR, Low DE): These are the key driving factors of the system. They form the foundation of the structure and must be prioritized in decision-making and strategic planning.

Together, ISM and MICMAC provide a powerful framework for understanding, modeling, and analyzing complex decision environments.

4.3. A New Approach for Risk Analysis: Integration of RPN and MICMAC

4.3.1. RPN – Risk Priority

The Risk Priority Number (RPN) method is a widely used quantitative risk assessment and prioritization technique for ranking risks according to their relative importance. Originating from the Failure Mode and Effects Analysis (FMEA) framework, RPN provides a systematic approach to identifying which risks require priority attention by combining multiple dimensions of risk into a single numerical indicator. This method supports the decision-making process by enabling clear and consistent prioritization of risks.

The RPN methodology evaluates each risk according to three key criteria: Severity (S), Occurrence (O), and Detection (D).

Severity represents the magnitude of the negative impact if the risk occurs and is evaluated on a scale from low to high; higher values indicate more serious consequences. Occurrence reflects the probability of the risk occurring and is similarly evaluated from low to high; higher values correspond to a higher probability of occurrence.

Detectability refers to the ability to identify or prevent the risk before it leads to negative impacts and is evaluated on an inverse logic scale; Lower values indicate higher

detectability, while higher values indicate lower detectability, meaning the risk is more difficult to detect or control.

Each criterion is scored using a predefined rank scale, usually ranging from 1 (lowest) to 10 (highest), based on expert opinion or available empirical evidence. Accordingly, higher Severity and Occurrence values increase the criticality of the risk, while higher Detection values indicate weaker detection ability and therefore higher risk importance.

For each risk, the Risk Priority Number (RPN) is obtained by multiplying severity, occurrence, and detectability, as defined in (10).

$$RPN = S \times O \times D \quad (10)$$

Due to this multiplicative structure, an increase in any one of the three components results in a higher RPN value, signaling greater overall risk priority. Once calculated, RPN values are ranked from highest to lowest, allowing risks with the greatest criticality to be identified and addressed first. Preventive or corrective actions are typically proposed for risks with high RPN scores, after which the values of S, O, and D may be reassessed to evaluate the effectiveness of the implemented measures

RPN analysis does not explicitly consider causal relationships or interdependencies among risks; instead, each risk is evaluated independently to determine its standalone priority level.

The main advantage of the RPN method lies in its simplicity and transparency, as it enables diverse risks to be compared within a unified numerical framework. However, because identical RPN values may result from different combinations of Severity, Occurrence, and Detection scores, the results should be interpreted carefully. Therefore, alongside the overall RPN ranking, examining the individual component scores is essential for identifying which dimension contributes most to the risk and for designing more effective risk mitigation strategies.

4.3.2. An Integrated Approach for RPN and MICMAC Analysis

In the ISM and MICMAC methodologies, risks are primarily evaluated based on their interrelationships and interaction structures within the system. These methods provide a robust analytical framework for identifying the hierarchical positions of risks as well as their driving and dependent roles. However, it is considered that the parameters used in the RPN method namely probability, impact (severity), and detectability offer valuable project specific insights that are not explicitly incorporated into ISM and MICMAC analyses. The

exclusion of these parameters in ISM and MICMAC based evaluations has therefore created the need to develop a new integrated approach that captures both relational and project dependent characteristics of risks.

In this study, RPN is incorporated into the MICMAC based structure with the aim of enriching the relational analysis of risks by embedding project specific risk attributes into the system oriented framework. While ISM and MICMAC define the structural positions of risks based on a fixed network of interactions, RPN reflects how critical a particular risk is within the context of a given project. Accordingly, RPN is not treated as an independent prioritization tool, but rather as a complementary component that enhances the structural assessment provided by MICMAC.

Within the proposed model, the driving power (DP) obtained from the MICMAC analysis is preserved as the primary indicator representing a risk's ability to influence other risks in the system. Since a risk that triggers or amplifies multiple other risks is considered structurally critical, the integration specifically focuses on the driving dimension. The dependence dimension, which reflects the extent to which a risk is influenced by others, is therefore not used as a prioritization criterion in this framework.

The inclusion of RPN within the MICMAC based structure enables the same set of risk factors to exhibit different levels of importance across different projects. Project scope, scale, technical complexity, and implementation conditions can significantly alter both the likelihood of a risk occurring and its potential impact on the project. Such variability cannot be captured through ISM and MICMAC alone, as the relational structure remains constant and intrinsic risk characteristics are not explicitly considered. By integrating RPN, the same risks can assume different probability and impact values depending on the project context, thereby introducing context sensitivity and diversity into the analysis even when the risk set remains unchanged.

To ensure comparability across projects with different numbers of risk factors, the driving power is normalized by dividing it by the total number of risks. The integrated risk priority score is then calculated by weighting the conventional RPN with the normalized driving power, as defined in (11).

$$RPN_{DP} = RPN \times \frac{DP}{n} \quad (11)$$

where RPN represents the project specific individual importance of the risk based on probability, impact, and detectability; DP denotes the driving power obtained from MICMAC analysis; and n indicates the total number of risk factors included in the analysis.

The use of a standardized scoring scale is essential, as the number of risks considered may vary significantly from one project to another. Without normalization, the resulting priority scores would lie on different scales, making consistent interpretation and comparison across projects infeasible. Through normalization, the RPN_{DP} value becomes a project independent, comparable, and interpretable risk priority indicator.

In conclusion, the integration of RPN into a MICMAC based framework provides a comprehensive analytical structure that simultaneously evaluates both the structural roles of risks within the system and their project specific critical characteristics. This integrated approach enables more realistic prioritization by accounting for both a risk's triggering influence on other risks and its probability, impact, and detectability within a given project. The resulting RPN_{DP} values serve as the basis of a quantitative decision support system, objectively identifying which risks require more urgent and strategic intervention and supporting users in developing response and action plans in a systematic, comparable, and project sensitive manner.

5. IMPLEMENTATION AND RESULTS

5.1. Data Collected

The dataset used in this study was obtained through interviews with the Integrated Project Management company. It consists of risk analyses prepared for 10 different construction projects previously undertaken by the company, which serve as the primary data source. These risk analyses contain qualitative information such as risk descriptions, potential outcomes, whether the risks are controllable, and the disciplines and sub-disciplines to which the risks are related. These risk analyses from various projects were consolidated into a single database, resulting in a raw dataset of 479 rows.

The combined risk dataset was reviewed. Repeating risks, those related to similar disciplines, and risks leading to similar outcomes were identified. The frequency of occurrence, the disciplines and sub-disciplines they belong to, and the processes they impact were analyzed together. Risks of similar nature were grouped under 20 "possible scenarios." To create a more consistent classification, the lifecycle of the construction projects was divided into five main processes: Initiation, Planning, Execution, Monitoring, and Closure. Each possible scenario was assigned to the relevant process or processes. Furthermore, the identified possible scenarios were classified into five main risk categories—external, financial, operational, project management, and technical risks—this classification is schematically presented in Figure 3.

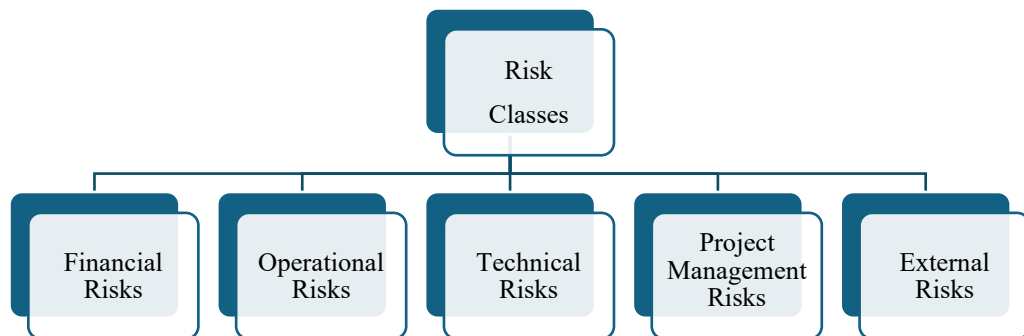


Figure 3. Classification of potential risk

To define the scope of each possible scenario more precisely, explanatory texts describing the corresponding risk clusters were developed. These descriptions were validated by referencing the expressions present in the actual risk data. Furthermore, for each possible scenario, verbatim examples directly extracted from the raw data were utilized, thereby

demonstrating that the defined clusters are firmly grounded in field-based and project-specific evidence. Throughout this process, the generated descriptions and selected examples were reviewed by subject-matter experts within the company, and revisions were implemented in line with the feedback received. As a result, both the theoretical coherence and the practical validity of the risk classification were reinforced through expert evaluation.

The 20 possible scenarios identified in this study, along with their associated risk classes, are presented in Table 2.

Table 2. Definition and categorization of project risk scenarios

ID	Category	Risk Description
R1	Financial	Financial and economic fluctuations
R2	Financial	Financial condition problems
R3	Operational	Site organization and coordination problems
R4	Operational	Insufficient personnel and lack of expertise
R5	Operational	Material supply delays
R6	Operational	Equipment and machinery supply delays
R7	Operational	Insufficient storage and working areas
R8	Operational	Technical problems caused by site and terrain conditions
R9	Operational	Occupational health and safety problems
R10	Project Management	Scheduling and work program errors
R11	Project Management	Errors arising from cost estimation
R12	Project Management	Scope creep
R13	Technical	Errors originating from project documents and drawings
R14	Technical	Design and engineering errors
R15	Technical	Errors arising from manufacturing, production, and installation
R16	Technical	Measurement, inspection, and verification errors
R17	External	Delays in approval processes
R18	External	Adverse weather conditions
R19	External	Employer-related issues
R20	External	Uncontrollable external factors

In this study, the identified risk scenarios were elaborated based on the descriptions and examples found in the raw risk data.

Financial and economic fluctuations (R1) refer to situations in which exchange rate volatility, inflation, and economic uncertainty in the market lead to unforeseen increases in

procurement, labor, and overall project costs. These effects may result in budget overruns. Examples such as “An additional payment is requested due to the increased steel quantity in accordance with the provisions stated in the unit price description for Steel Structure Manufacturing” and “Costs exceeding forecasts due to an unforeseen increase in the exchange rate during the design process” support this risk scenario.

Financial condition problems (R2) describe financial difficulties arising from poor financial management by the contractor. Such situations may lead to project suspension and/or contractor bankruptcy, resulting in withdrawal from the project. Examples identified in the raw data include “Failure to receive payments due to the employer’s financial difficulties or bankruptcy,” “Failure to reach the targeted level due to rejected credit applications,” and “The contractor declaring concordat.”

Site organization and coordination problems (R3) include situations where a lack of coordination between teams, incorrect guidance, or managerial deficiencies slow down workflow. These issues cause clashes and productivity losses on site. Examples such as “Lack of coordination with other disciplines” and “Schedule delays resulting from clashes between electrical and mechanical piping systems” illustrate this risk.

Insufficient personnel and lack of expertise (R4) cover situations where untrained, inexperienced, or inadequate personnel reduce work quality and cause defective workmanship or occupational accidents. Examples include “Restoration works not being supervised by a specialized restorer,” “Damage to completed works by inexperienced personnel during execution,” and “Non-compliant workmanship.”

Material supply delays (R5) refer to disruptions in the work schedule due to the inability to procure materials and equipment on time, leading to delays in dependent activities. Examples such as “All ongoing tendering and material procurement processes being suspended by the main contractor” and “Chip crisis” were recorded under this risk category.

Equipment and machinery supply delays (R6) involve situations where critical equipment cannot be delivered on time, preventing the initiation of dependent site activities. Examples include “Testing and commissioning processes dependent on equipment” and “Delays in the procurement of fire detection and alarm systems.”

Insufficient storage and working areas (R7) describe situations where limited site space complicates coordination among teams and reduces operational efficiency. Examples include “Risk of occupational accidents in areas where work is carried out alongside neighboring stakeholders,” “Excavation works progressing slower than expected due to limited site space,” and “Delays in material logistics due to the lack of storage and working areas on site.”

Technical problems caused by site and terrain conditions (R8) encompass issues arising from land structure, soil conditions, existing buildings, or unforeseen site-specific conditions encountered during construction. Examples include “Ventilation problems due to well depth” and “Displacement occurring in the existing retaining system of neighboring parcels.”

Occupational health and safety problems (R9) refer to work stoppages and occupational accidents caused by insufficient OHS inspections, failure to use personal protective equipment, or hazardous working environments. Examples include “Decrease in OHS inspections due to a reduction in OHS personnel” and “Occupational accidents occurring in areas where buildings are closely spaced.”

Scheduling and work program errors (R10) include incorrect duration estimates, sequencing errors, or unrealistic scheduling decisions that negatively affect overall project progress. Examples such as “Design+Build scaffolding installation taking longer than planned” and “Schedule delays caused by clashes between electrical and mechanical piping systems” support this risk.

Errors arising from cost estimation (R11) refer to unrealistic budgets caused by incorrect quantity take-offs, inappropriate unit price selections, overlooked work items, or misjudgment of market conditions during the budgeting phase. Incomplete concrete and reinforcement quantity calculations, errors in unit price analyses, and missing work items were evaluated within this scope.

Scope creep (R12) involves situations where changes, revisions, or additional requirements are requested by the employer, consultant, or user after completion of works, necessitating rework or additional construction. Examples include requests to change the color of completed wall cladding, modification of the route of finished mechanical lines, functional

changes requested by users after delivery, and subsequent alterations to finalized design details.

Errors originating from project documents and drawings (R13) include incomplete, incorrect, or revision-prone drawings that cause implementation errors, work stoppages, and rework. Issues identified between the 21st and 27th floors of Tower Parcel 12, increases in quantities and construction durations due to project revisions, and the completion of the Design+Build scaffolding final project with greater quantities than expected support this risk.

Design and engineering errors (R14) refer to technical problems arising during implementation due to insufficient calculations, incompatible engineering decisions, or design gaps. Examples include “The designer failing to complete the work on time” and “Incomplete power calculations.”

Errors arising from manufacturing, production, and installation (R15) encompass situations requiring rework and affecting the schedule due to incorrect execution, out-of-tolerance manufacturing, or installation errors. Examples include door and joinery dimension errors, damage to completed works, damage to historic doors and joinery during dismantling, errors in insulation and waterproofing details, and applications carried out under unsuitable temperature and humidity conditions.

Measurement, inspection, and verification errors (R16) include problems encountered during testing, adjustment, and commissioning phases of mechanical, electrical, and automation systems that extend project delivery or result in performance deficiencies. Errors in fixed furniture measurements, commissioning delays, delays in system integration, and inaccurate excavation area measurements were evaluated under this category.

Delays in approval processes (R17) refer to situations where delays in approvals from employers, municipalities, or similar authorities slow down workflow and prevent the initiation of critical activities. Delays in UTK approval processes and potential issues in municipal permitting procedures are examples of this risk.

Adverse weather conditions (R18) include situations where severe wind, rain, snow, or low temperatures halt activities such as façade works, reinforced concrete operations, and crane

usage. Non-working days due to unfavorable weather and completed works being exposed to harsh weather conditions support this risk.

Employer-related issues (R19) encompass situations where disputes between parties directly affect project flow, leading to work stoppages, claims, or delays. The financial problems of the main contractor in a defense industry project carried out in Ankara were evaluated within this scope.

Finally, uncontrollable external factors (R20) refer to events beyond control—such as earthquakes, pandemics, epidemics, or governmental decisions—that directly affect project activities. The COVID-19 pandemic is cited as a representative example of this risk scenario.

5.2. Implementation of ISM

5.2.1. Establishing Contextual Links Between Factors

In order to identify the relationships among the risk factors, independent opinions were obtained from three different experts. The experts were asked the question, “Does the occurrence of Risk 1 influence the occurrence of Risk 2?” for each pair of risk factors, and this evaluation was conducted separately for all possible risk pairs. The expert judgments obtained were first assessed based on the principle of majority voting, and the Structural Self-Interaction Matrix (SSIM) was constructed accordingly. In cases where all three experts provided differing opinions, the assessment of the expert with the highest level of professional experience was adopted. Within this framework, the experts were ranked in terms of experience as follows: Senior Architect, Senior Design Manager, and Technical Office Engineer. The SSIM matrix developed based on expert judgments is presented below. Table 3 illustrates the final form of the SSIM matrix.

Table 3. SSIM matrix

	R20	R19	R18	R17	R16	R15	R14	R13	R12	R11	R10	R9	R8	R7	R6	R5	R4	R3	R2	R1
R1	A	O	O	V	O	O	O	O	V	V	V	O	O	O	V	V	V	O	V	
R2	A	X	O	V	O	O	O	O	V	V	V	O	O	O	V	V	V	V		
R3	O	X	O	O	O	O	O	O	O	V	V	V	A	A	V	V	V			
R4	O	X	O	V	V	V	O	V	O	V	V	V	O	O	V	V				
R5	A	O	A	X	O	O	A	O	V	V	V	O	O	A	O					
R6	A	O	A	X	O	O	A	O	V	V	V	O	O	A						
R7	O	O	O	O	O	O	O	O	O	O	O	V	A							
R8	O	O	O	O	O	V	O	O	V	V	V	V								
R9	O	O	O	O	O	A	O	O	O	O	O									
R10	A	O	A	V	A	A	A	A	A	X										
R11	A	O	A	V	A	A	A	A	X											
R12	O	A	O	V	O	A	A	A												
R13	O	O	O	O	O	V	A													
R14	O	O	O	O	V	V														
R15	O	O	A	O	V															
R16	O	O	A	O																
R17	O	A	A																	
R18	A	O																		
R19	O																			
R20																				

5.2.2. Transformation of the SSIM into the Initial Reachability Matrix (T)

In order to transform the qualitatively identified relationships obtained through the Structural Self-Interaction Matrix (SSIM) into a quantitative form, the SSIM was converted into a binary adjacency matrix. This transformation enabled the interactions among risk factors to be represented within a binary (0–1) structure, thereby providing the necessary input for the subsequent stages of the ISM analysis.

Based on the standard transformation rules defined in the ISM methodology, the symbols used in the SSIM were systematically converted into binary values. Accordingly, when one risk factor was determined to influence another, the corresponding matrix element was assigned a value of “1”, whereas a value of “0” was assigned in the absence of any influence. Depending on whether the SSIM symbols indicated a directional or mutual relationship, the transformation process was consistently applied to both the a_{ij} and a_{ji} elements.

As a result of this process, the adjacency matrix (A), representing only the direct relationships among the risk factors, was obtained, as presented in Table 4.

Table 4. Adjacency matrix (A)

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
R1		1	0	1	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R2	0		1	1	1	1	0	0	0	1	1	1	0	0	0	0	1	0	1	0
R3	0	0		1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	1	0
R4	0	0	0		1	1	0	0	1	1	1	0	1	0	1	1	1	0	1	0
R5	0	0	0	0		0	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R6	0	0	0	0	0		0	0	0	1	1	1	0	0	0	0	1	0	0	0
R7	0	0	1	0	1	1		0	1	0	0	0	0	0	0	0	0	0	0	0
R8	0	0	1	0	0	0	1		1	1	1	1	0	0	1	0	0	0	0	0
R9	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0
R10	0	0	0	0	0	0	0	0	0		1	0	0	0	0	0	1	0	0	0
R11	0	0	0	0	0	0	0	0	0	1		1	0	0	0	0	1	0	0	0
R12	0	0	0	0	0	0	0	0	0	1	1		0	0	0	0	1	0	0	0
R13	0	0	0	0	0	0	0	0	0	1	1	1		0	1	0	0	0	0	0
R14	0	0	0	0	1	1	0	0	0	1	1	1	1		1	1	0	0	0	0
R15	0	0	0	0	0	0	0	0	1	1	1	1	0	0		1	0	0	0	0
R16	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0		0	0	0	0
R17	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0		0	0	0
R18	0	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	1		0	0
R19	0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0		0
R20	1	1	0	0	1	1	0	0	0	1	1	0	0	0	0	0	0	1	0	

Since self-relationships are not explicitly defined within the adjacency matrix, an identity matrix was subsequently added to incorporate self-reachability into the system. This operation ensures that each risk factor is assumed to be reachable from itself, which is a fundamental requirement in the ISM framework. Following this step, the Initial Reachability Matrix (T) was obtained, as shown in Table 5.

Table 5. Initial reachability matrix (T)

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
R1	1	1	0	1	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R2	0	1	1	1	1	1	0	0	0	1	1	1	0	0	0	0	1	0	1	0
R3	0	0	1	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	1	0
R4	0	0	0	1	1	1	0	0	1	1	1	0	1	0	1	1	1	0	1	0
R5	0	0	0	0	1	0	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R6	0	0	0	0	0	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R7	0	0	1	0	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0
R8	0	0	1	0	0	0	1	1	1	1	1	1	0	0	1	0	0	0	0	0
R9	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
R10	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0
R11	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R12	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R13	0	0	0	0	0	0	0	0	0	1	1	1	1	0	1	0	0	0	0	0
R14	0	0	0	0	1	1	0	0	0	1	1	1	1	1	1	1	0	0	0	0
R15	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	0	0	0	0
R16	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	1	0	0	0	0
R17	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0
R18	0	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	1	1	0	0
R19	0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1	0
R20	1	1	0	0	1	1	0	0	0	1	1	0	0	0	0	0	0	1	0	1

This matrix serves as the basis for the transitivity analysis and the level partitioning procedures carried out in the subsequent stages of the ISM implementation.

5.2.3. Evaluation of Transitive Relationships and Generation of the Final Reachability Matrix (T^*)

The initial reachability matrix (T) was processed in accordance with the transitivity assumption, which is a fundamental principle of Interpretive Structural Modeling (ISM). This assumption states that if factor i influences factor j , and factor j influences factor k , then factor i is also assumed to have an indirect influence on factor k . To capture these indirect relationships, the initial reachability matrix was analyzed using Boolean algebra.

In the first step, the identity matrix (I) was added to the initial reachability matrix to represent the self-reachability of each factor, resulting in the matrix T^1 . The matrix was then iteratively processed using Boolean matrix multiplication. At each iteration, a new matrix T^{k+1} was

obtained and compared with the matrix from the previous iteration T^k . This iterative procedure continued until convergence was achieved, that is, until no changes were observed between two successive matrices ($T^{k+1} = T^k$).

The converged matrix, denoted as T^k , represents the final reachability matrix incorporating all direct and indirect relationships among the factors. This matrix served as the basis for the subsequent level partitioning stage of the ISM analysis.

Iteration 1

The matrix T^2 was computed from T using the Boolean operations described in the Table 6.

Table 6. T^2 matrix

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
R1	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R2	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R3	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R4	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R5	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R6	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R7	0	0	1	1	1	1	1	0	1	1	1	1	0	0	0	0	1	0	1	0
R8	0	0	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	0	1	0
R9	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
R10	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R11	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R12	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R13	0	0	0	0	0	0	0	0	1	1	1	1	1	0	1	1	1	0	0	0
R14	0	0	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	0	0	0
R15	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	1	0	0	0
R16	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	1	1	0	0	0
R17	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R18	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0
R19	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R20	1	1	1	1	1	1	0	0	0	1	1	1	0	0	1	1	1	1	1	1

Table 6 presents the matrix T^2 . Since $T^2 \neq T^1$, further iterations were required and the proceeded to the next step.

Iteration 2

Using T^2 as the input, the same Boolean operation was repeated to derive T^3 .

Table 7. T^3 matrix

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
R1	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R2	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R3	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R4	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R5	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R6	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R7	0	1	1	1	1	1	1	0	1	1	1	1	1	0	1	1	1	0	1	0
R8	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0
R9	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
R10	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R11	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R12	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R13	0	0	0	0	1	1	0	0	1	1	1	1	1	0	1	1	1	0	0	0
R14	0	0	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	0	0	0
R15	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	0	0	0
R16	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	0
R17	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R18	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0
R19	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R20	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	1	1	1

Table 7 presents the matrix T^3 . As $T^3 \neq T^2$, further iterations were required and the process proceeded to the next step.

Iteration 3

In the subsequent step, T^4 was computed based on T^3 following the same iterative rule.

Table 8. T^4 matrix

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
R1	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R2	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R3	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R4	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R5	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R6	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R7	0	1	1	1	1	1	1	0	1	1	1	1	1	0	1	1	1	0	1	0
R8	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0
R9	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
R10	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R11	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R12	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R13	0	0	0	0	1	1	0	0	1	1	1	1	1	0	1	1	1	0	0	0
R14	0	0	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	0	0	0
R15	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	0	0	0
R16	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	0
R17	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0
R18	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0
R19	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0
R20	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	1	1	1

Table 8 shows the T^4 matrix. Since $T^4 = T^3$ was satisfied, the matrix considered stable and accepted as the final reachability matrix T^* .

5.2.4. Calculation of Reachability, Antecedent, and Intersection Sets

Table 9. Reachability, antecedent, and intersection sets

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)
R1	{1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 20}	{1}
R2	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}
R3	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}
R4	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}
R5	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}
R6	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}
R7	{2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{7, 8}	{7}
R8	{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{8}	{8}
R9	{9}	{1, 2, 3, 4, 7, 8, 9, 13, 14, 15, 18, 19, 20}	{9}
R10	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}
R11	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}
R12	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}
R13	{5, 6, 9, 10, 11, 12, 13, 15, 16, 17}	{1, 2, 3, 4, 7, 8, 13, 14, 19, 20}	{13}
R14	{5, 6, 9, 10, 11, 12, 13, 14, 15, 16, 17}	{14}	{14}
R15	{5, 6, 9, 10, 11, 12, 15, 16, 17}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 18, 19, 20}	{15}
R16	{5, 6, 10, 11, 12, 16, 17}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 16, 18, 19, 20}	{16}
R17	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}
R18	{5, 6, 9, 10, 11, 12, 15, 16, 17, 18}	{18, 20}	{18}
R19	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}
R20	{1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20}	{20}	{20}

After obtaining the final reachability matrix T^* , the relative position of each risk within the system was examined through the calculation of reachability, antecedent, and intersection sets. This step serves as an application-oriented transition from the matrix-based representation of relationships to a form that enables hierarchical interpretation.

For each risk R_i , the reachability set was constructed by identifying all risks corresponding to the columns with a value of “1” in the i -th row of the final reachability matrix. This set represents the risks that can be reached from R_i , including both direct and indirect relationships. Conversely, the antecedent set was obtained by identifying all risks corresponding to the rows with a value of “1” in the i -th column of the matrix, indicating the risks that influence R_i . The intersection set was then formed as the common subset of the reachability and antecedent sets for each risk.

Table 9 presents the reachability, antecedent, and intersection sets derived for all risks based on the final reachability matrix T^* .

5.2.5. Level Partitioning and Hierarchical Diagram

Following the determination of the reachability, antecedent, and intersection sets, the level partitioning procedure was applied in accordance with the ISM methodology. A risk was assigned to a given hierarchical level when its reachability set was identical to its intersection set ($C(i) = R(i)$), indicating that the risk does not lead to any other risks beyond those with which it shares mutual influence.

Risks satisfying this condition were fixed at the corresponding level and subsequently removed from further iterations. The same procedure was then repeatedly applied to the reduced matrix until all risks were assigned to a hierarchical level. Through this iterative elimination process, a clear and structured hierarchy of risks was established.

Iteration 1

Table 10 shows the first iteration.

Table 10. First iteration

	Reachability Set (R _i)	Antecedent Set (A _i)	Intersection Set (C _i)	Levels
R1	{1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 20}	{1}	
R2	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R3	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R4	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R5	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}	Level 1
R6	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}	Level 1
R7	{2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{7, 8}	{7}	
R8	{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{8}	{8}	
R9	{9}	{1, 2, 3, 4, 7, 8, 9, 13, 14, 15, 18, 19, 20}	{9}	Level 1
R10	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}	Level 1
R11	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}	Level 1
R12	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}	Level 1
R13	{5, 6, 9, 10, 11, 12, 13, 15, 16, 17}	{1, 2, 3, 4, 7, 8, 13, 14, 19, 20}	{13}	
R14	{5, 6, 9, 10, 11, 12, 13, 14, 15, 16, 17}	{14}	{14}	
R15	{5, 6, 9, 10, 11, 12, 15, 16, 17}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 18, 19, 20}	{15}	
R16	{5, 6, 10, 11, 12, 16, 17}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 16, 18, 19, 20}	{16}	
R17	{5, 6, 10, 11, 12, 17}	{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}	{5, 6, 10, 11, 12, 17}	Level 1
R18	{5, 6, 9, 10, 11, 12, 15, 16, 17, 18}	{18, 20}	{18}	
R19	{2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R20	{1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20}	{20}	{20}	

Since $C(i) = R(i)$ in R5, R6, R9, R10, R11 and R12 are placed in Level 1.

Iteration 2

Table 11 shows the second iteration. After the first iteration, R5, R6, R9, R10, R11, R12 and R17 were removed from the system.

Table 11. Second iteration

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)	Levels
R1	{1, 2, 3, 4, 13, 15, 16, 19}	{1, 20}	{1}	
R2	{2, 3, 4, 13, 15, 16, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R3	{2, 3, 4, 13, 15, 16, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R4	{2, 3, 4, 13, 15, 16, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R7	{2, 3, 4, 7, 13, 15, 16, 19}	{7, 8}	{7}	
R8	{2, 3, 4, 7, 8, 13, 15, 16, 19}	{8}	{8}	
R13	{13, 15, 16}	{1, 2, 3, 4, 7, 8, 13, 14, 19, 20}	{13}	
R14	{13, 14, 15, 16}	{14}	{14}	
R15	{15, 16}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 18, 19, 20}	{15}	
R16	{16}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 16, 18, 19, 20}	{16}	Level 2
R18	{15, 16, 18}	{18, 20}	{18}	
R19	{2, 3, 4, 13, 15, 16, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R20	{1, 2, 3, 4, 13, 15, 16, 18, 19, 20}	{20}	{20}	

Since $C(i) = R(i)$ in R16 is placed in Level 2.

Iteration 3

Table 12 shows the third iteration. After the second iteration, R16 was removed from the system.

Table 12. Third iteration

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)	Levels
R1	{1, 2, 3, 4, 13, 15, 19}	{1, 20}	{1}	
R2	{2, 3, 4, 13, 15, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R3	{2, 3, 4, 13, 15, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R4	{2, 3, 4, 13, 15, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R7	{2, 3, 4, 7, 13, 15, 19}	{7, 8}	{7}	
R8	{2, 3, 4, 7, 8, 13, 15, 19}	{8}	{8}	
R13	{13, 15}	{1, 2, 3, 4, 7, 8, 13, 14, 19, 20}	{13}	
R14	{13, 14, 15}	{14}	{14}	
R15	{15}	{1, 2, 3, 4, 7, 8, 13, 14, 15, 18, 19, 20}	{15}	Level 3
R18	{15, 18}	{18, 20}	{18}	
R19	{2, 3, 4, 13, 15, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R20	{1, 2, 3, 4, 13, 15, 18, 19, 20}	{20}	{20}	

Since $C(i) = R(i)$ in R15 is placed in Level 3.

Iteration 4

Table 13 shows the fourth iteration. After the third iteration, R15 was removed from the system.

Table 13. Fourth iteration

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)	Levels
R1	{1, 2, 3, 4, 13, 19}	{1, 20}	{1}	
R2	{2, 3, 4, 13, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R3	{2, 3, 4, 13, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R4	{2, 3, 4, 13, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R7	{2, 3, 4, 7, 13, 19}	{7, 8}	{7}	
R8	{2, 3, 4, 7, 8, 13, 19}	{8}	{8}	
R13	{13}	{1, 2, 3, 4, 7, 8, 13, 14, 19, 20}	{13}	Level 4
R14	{13, 14}	{14}	{14}	
R18	{18}	{18, 20}	{18}	Level 4
R19	{2, 3, 4, 13, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	
R20	{1, 2, 3, 4, 13, 18, 19, 20}	{20}	{20}	

Since $C(i) = R(i)$ in R13 and R18 are placed in Level 4.

Iteration 5

Table 14 shows the fifth iteration. After the fourth iteration, R13 and R18 were removed from the system.

Table 14. Fifth iteration

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)	Levels
R1	{1, 2, 3, 4, 19}	{1, 20}	{1}	
R2	{2, 3, 4, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	Level 5
R3	{2, 3, 4, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	Level 5
R4	{2, 3, 4, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	Level 5
R7	{2, 3, 4, 7, 19}	{7, 8}	{7}	
R8	{2, 3, 4, 7, 8, 19}	{8}	{8}	
R14	{14}	{14}	{14}	Level 5
R19	{2, 3, 4, 19}	{1, 2, 3, 4, 7, 8, 19, 20}	{2, 3, 4, 19}	Level 5
R20	{1, 2, 3, 4, 19, 20}	{20}	{20}	

Since $C(i) = R(i)$ in R2, R3, R4, R14 and R19 are placed in Level 5.

Iteration 6

Table 15 shows the sixth iteration. After the fifth iteration, R2, R3, R4, R14 and R19 were removed from the system.

Table 15. Sixth iteration

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)	Levels
R1	{1}	{1, 20}	{1}	Level 6
R7	{7}	{7, 8}	{7}	Level 6
R8	{7, 8}	{8}	{8}	
R20	{1, 20}	{20}	{20}	

Since $C(i) = R(i)$ in R1 and R7 are placed in Level 6.

Iteration 7

Table 16 shows the seventh iteration. After the sixth iteration, R1 and R7 were removed from the system.

Table 16. Seventh iteration

	Reachability Set (R_i)	Antecedent Set (A_i)	Intersection Set (C_i)	Levels
R8	{8}	{8}	{8}	Level 7
R20	{20}	{20}	{20}	Level 7

Since $C(i) = R(i)$ in R8 and R20 are placed in Level 7.

Table 17 presents the finalized level structure and the risks assigned to each level.

Table 17. Level partitioning

Level	Risk Factors
Level 1	R5, R6, R9, R10, R11, R12, R17
Level 2	R16
Level 3	R15
Level 4	R13, R18
Level 5	R2, R3, R4, R14, R19
Level 6	R1, R7
Level 7	R8, R20

Following the completion of the level partitioning procedure, the hierarchical structure of the risk system was transformed into a directed Interpretive Structural Modeling (ISM) diagram. The levels identified in Table Y were directly used to position each risk within the diagram, ensuring a clear correspondence between the analytical results and their visual representation.

Level 1 risks, which represent outcome-oriented and highly dependent elements, were placed at the top of the diagram, while risks with higher driving influence were positioned progressively at lower levels. This vertical arrangement reflects the directional flow of influence within the system.

The links between risks were established based on the final reachability matrix T^* . To improve clarity, only direct relationships were included, and transitive links were removed, as indirect effects are already captured through the hierarchical structure. The resulting ISM diagram corresponding to the identified level structure is presented in Figure 4.

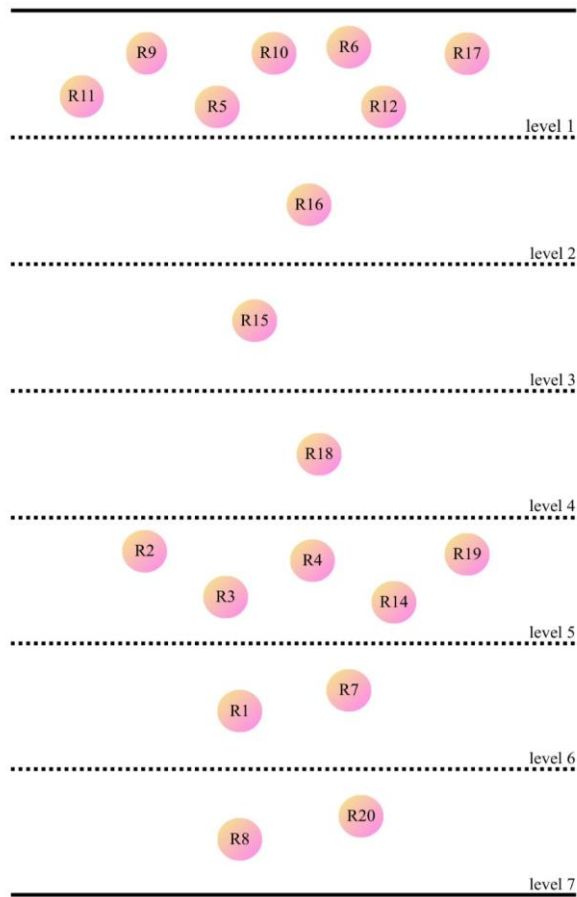


Figure 4. ISM diagram

5.3. Implementation of MICMAC

Following the completion of the ISM phase, the MICMAC analysis is conducted using the final reachability matrix derived as the output of the ISM procedure. This matrix encapsulates both direct and indirect relationships among the identified risk factors, thereby representing the stabilized structural configuration of the risk system. As such, it constitutes the primary analytical input for evaluating the influence–dependence characteristics of the risks.

Table 18. Final reachability matrix DR&DP

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20	DRI
R1	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0	15
R2	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0	14
R3	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0	14
R4	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0	14
R5	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	6
R6	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	6
R7	0	1	1	1	1	1	1	0	1	1	1	1	1	0	1	1	1	0	1	0	15
R8	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0	16
R9	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1
R10	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	6
R11	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	6
R12	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	6
R13	0	0	0	0	1	1	0	0	1	1	1	1	1	0	1	1	1	0	0	0	10
R14	0	0	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	0	0	0	11
R15	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	0	0	0	9
R16	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	0	7
R17	0	0	0	0	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	6
R18	0	0	0	0	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0	10
R19	0	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	0	1	0	14
R20	1	1	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	1	1	1	17
DPj	2	8	8	8	19	19	2	1	13	19	19	19	10	1	12	13	19	2	8	1	

Based on the Table 18, the driving power (DR_i) and dependence power (DP_i) of each risk factor are calculated through row-wise and column-wise summations, respectively. The driving power reflects the extent to which a given risk influences other risks within the system, whereas the dependence power indicates the degree to which a risk is influenced by others. These quantitative measures enable a systematic classification of risks and form the basis for the subsequent MICMAC categorization and graphical representation.

An examination of the calculated values reveals that R20, R8, and R1/R7 exhibit the highest driving power levels within the system. In particular, R20 demonstrates the strongest driving role, indicating that it acts as a dominant source of influence capable of triggering a wide range of other risks. Similarly, R8 shows a high propagation potential, suggesting its critical

position in initiating cascading effects across the risk network. R1 and R7, which share comparable driving power values, can be interpreted as key leverage risks that significantly shape the overall behavior of the system.

In terms of dependence power, several risks, including R5, R6, and R10 display the highest DP_i values; reflecting a strong reliance on other risks within the system. These risks are predominantly outcome-oriented, as their activation is largely contingent upon the occurrence of multiple upstream risks. Their high dependence levels suggest that effective management of driving risks may substantially mitigate the likelihood and impact of these dependent risks.

5.3.1. Graphical Representation of Risk Interactions via MICMAC Analysis

Following the calculation of the driving power (DR_i) and dependence power (DP_i) values, the MICMAC plot is constructed to visualize the structural position of each risk within the system. In this representation, the vertical axis denotes driving power, while the horizontal axis denotes dependence power, allowing risks to be positioned according to their relative influence and dependency characteristics. The coordinates of each risk are directly derived from the final reachability matrix, ensuring consistency between the ISM and MICMAC stages.

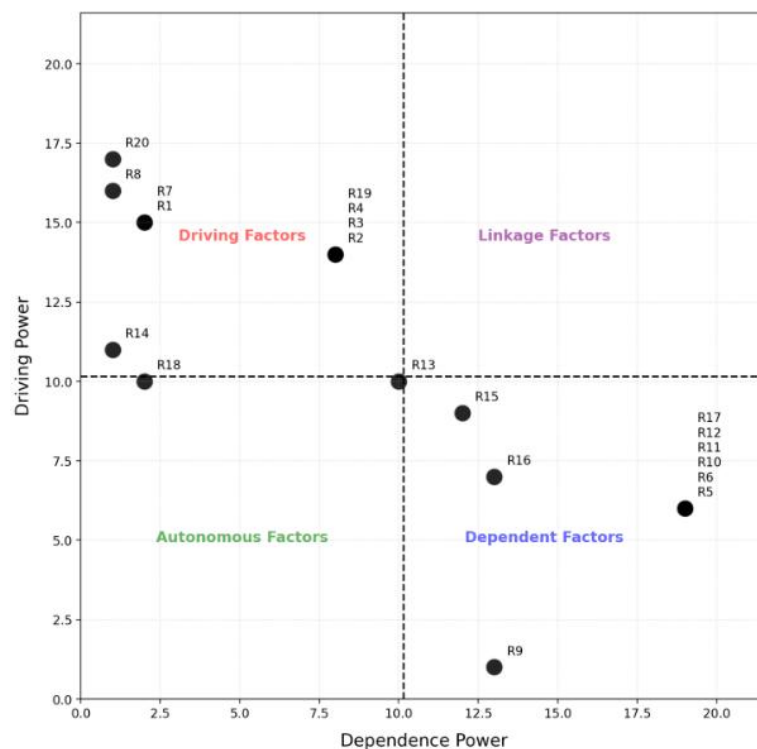


Figure 5. Graph of the MICMAC diagram

As illustrated in the Figure 5, the spatial distribution of risks highlights clear differences in their systemic roles. Risks located at higher vertical positions exhibit stronger driving effects, indicating a greater capacity to influence other risks, whereas risks positioned further along the horizontal axis reflect higher dependence on the overall system dynamics. This visual configuration provides an intuitive overview of how risks interact within the system and offers preliminary insights into which risks may act as key influencers or outcome-oriented elements. These observations form the basis for the subsequent classification and detailed interpretation of risks within the MICMAC framework.

5.3.2. Classification of Risk Factors into Four Categories

Based on the driving power (DR_i) and dependence power (DP_i) values obtained from the MICMAC analysis, the risk factors are classified into four distinct categories. This classification is performed according to the relative positioning of risks within the influence–dependence space and enables a systematic interpretation of their functional roles in the overall risk system. The categorization provides a simplified yet meaningful framework for understanding how different risks contribute to or are affected by system dynamics.

The classification results are summarized in Table 19, which presents the allocation of each risk factor into one of the four MICMAC categories based on their corresponding DR_i and DP_i values.

Table 19. Categories of the risk factors

Risk Categories	Risk Factors
Driving Factors	R20, R8 , R7, R1, R19, R4, R3, R2, R14
Linkage Factors	-
Autonomous Factors	R18, R13
Dependent Factors	R17, R12, R11, R10, R6, R5, R9, R16, R15

Based on the MICMAC analysis results summarized in Table 19, the risk factors are classified into four categories according to their driving and dependence power characteristics. The driving factors exhibit high driving power and relatively low dependence, indicating their dominant influence on the overall risk system and their potential to trigger cascading effects across multiple risks. No risk factors are identified as

linkage factors, suggesting the absence of risks with simultaneously high driving and dependence power and implying a relatively stable interaction structure within the system. The autonomous factors display low driving and low dependence power, reflecting their limited interaction with other risks and marginal role in shaping system-wide dynamics. In contrast, the dependent factors are characterized by high dependence and low driving power, indicating that they primarily represent outcome-oriented risks whose occurrence is largely influenced by upstream driving factors rather than acting as sources of influence themselves.

5.4. Implementation of a New Approach for Risk Analysis: Integration of RPN & MICMAC

Although the same set of risk factors may be identified across different construction projects, the relative importance of these risks can vary significantly depending on project-specific characteristics. Factors such as project type, scale, contractual structure, stakeholder profile, and organizational context may alter how risks are perceived, how likely they are to occur, and how severe their impacts may be. For this reason, one of the main motivations for incorporating the Risk Priority Number (RPN) approach into the existing analysis is the belief that risk evaluation should be conducted in a more dynamic and project-oriented manner, rather than relying solely on a fixed structural representation of risks.

5.4.1. Graphical Representation of Risk Interactions via MICMAC Analysis

The RPN assessment was conducted using the same risk factors previously identified through the ISM–MICMAC analysis in order to reflect project-specific risk perceptions. Evaluations were obtained from a single company expert who had also contributed to the earlier stages of the study, based on a case-based bank headquarters building construction project selected to demonstrate the practical application of the proposed framework. The RPN parameters such as probability of occurrence, impact severity, and detectability were assessed for each risk factor using a numerical scale ranging from 1 to 10. The assessments were informed by the expert's professional experience, accumulated knowledge from comparable large-scale construction projects, and professional judgment.

Table 20 presents the risk prioritization results obtained using the conventional Risk Priority Number (RPN) method based on the evaluated probability of occurrence, impact severity,

and detectability scores. The calculated RPN values provide an initial, project-specific ranking of risks, reflecting the relative criticality of each risk factor when assessed independently of their interrelationships within the system.

According to the results, errors arising from cost estimation (R11) emerge as the most critical risk, exhibiting the highest RPN value, followed by financial and economic fluctuations (R1) and material supply delays (R5). These risks are characterized by a combination of high occurrence likelihood and severe potential impacts, indicating their significant influence on project performance. The prominence of these risks in the RPN-based ranking highlights the importance of cost-related uncertainties and supply chain reliability in large-scale construction projects and provides a baseline reference for further evaluation through integrated analytical approaches.

RPN-based prioritization and ISM–MICMAC analysis yield complementary yet distinct insights into project risk criticality. While the RPN results highlight risks such as errors arising from cost estimation (R11) and material supply delays (R5) as highly critical due to their elevated probability–severity–detectability combinations, the MICMAC analysis identifies uncontrollable external factors (R20), technical problems caused by site and terrain conditions (R8), and financial and economic fluctuations (R1) as dominant driving risks within the system.

Table 20. Risk priority number (RPN) for the risk factors

#	Risk Factors	Occurrence (1-10)	Severity (1-10)	Detectability (10-1)	RPN
R1	Financial and economic fluctuations	7	9	6	378
R2	Financial condition problems	6	9	6	324
R3	Site organization and coordination problems	7	8	5	280
R4	Insufficient personnel and lack of expertise	6	7	5	210
R5	Material supply delays	7	8	6	336
R6	Equipment and machinery supply delays	5	7	6	210
R7	Insufficient storage and working areas	6	6	4	144
R8	Technical problems caused by site and terrain conditions	5	8	7	280
R9	Occupational health and safety problems	4	9	6	216
R10	Scheduling and work program errors	6	8	4	192
R11	Errors arising from cost estimation	7	9	7	441
R12	Scope creep	7	8	6	336
R13	Errors originating from project documents and drawings	6	8	5	240
R14	Design and engineering errors	4	9	6	216
R15	Errors arising from manufacturing, production, and installation	6	7	4	168
R16	Measurement, inspection, and verification errors	5	7	5	175
R17	Delays in approval processes	7	8	6	336
R18	Adverse weather conditions	5	6	3	90
R19	Employer-related issues	6	8	6	288
R20	Uncontrollable external factors	2	10	9	180

Notably, R20 and R8 do not rank among the highest risks in the conventional RPN list, yet their high driving power indicates a strong capacity to trigger cascading effects across multiple risk factors. R1 represents a partial overlap between the two methods, appearing as both a high RPN risk and a strong driving factor. This divergence demonstrates that RPN captures risk severity and immediacy, whereas MICMAC reveals systemic influence and propagation potential. The differing yet complementary outputs of the two methods provide

a clear rationale for the proposed integrated RPN–MICMAC approach, which aims to jointly account for both quantitative risk criticality and structural impact within the project risk system.

5.4.2. Implementation of a New Approach: RPN and MICMAC Analysis

The integrated RPN–MICMAC approach is introduced to overcome the limitations of relying solely on conventional RPN-based prioritization, which evaluates risks independently and does not account for their systemic influence within the project structure. By incorporating driving power information derived from the MICMAC analysis, the proposed integration aims to balance risk criticality with structural impact, enabling a more comprehensive and decision-oriented prioritization framework.

The results obtained from this integrated approach are presented in Table 21, where conventional RPN values are combined with driving power measures to derive adjusted risk scores. Since the RPN values were determined within the scope of a case-based representative project, the resulting ranking reflects a project-specific prioritization rather than a universally fixed ordering. The integrated results indicate that risks such as financial and economic fluctuations (R1), technical problems caused by site and terrain conditions (R8), and uncontrollable external factors (R20) gain prominence due to their strong driving roles, even when their standalone RPN values are relatively moderate. Conversely, some risks with high RPN values but limited driving influence are assigned lower priority within the integrated framework. This shift highlights the contribution of the proposed integration in identifying risks that are not only severe but also capable of triggering cascading effects across the risk system.

Table 21. Integrated RPN–MICMAC risk prioritization results (RPN_{DR})

#	Risk Factors	RPN	Driving Power	RPN_{DR}
R1	Financial and economic fluctuations	378	15	283,5
R2	Financial condition problems	324	14	226,8
R3	Site organization and coordination problems	280	14	196
R4	Insufficient personnel and lack of expertise	210	14	147
R5	Material supply delays	336	6	100,8
R6	Equipment and machinery supply delays	210	6	63
R7	Insufficient storage and working areas	144	15	108
R8	Technical problems caused by site and terrain conditions	280	16	224
R9	Occupational health and safety problems	216	1	10,8
R10	Scheduling and work program errors	192	6	57,6
R11	Errors arising from cost estimation	441	6	132,3
R12	Scope creep	336	6	100,8
R13	Errors originating from project documents and drawings	240	10	120
R14	Design and engineering errors	216	11	118,8
R15	Errors arising from manufacturing, production, and installation	168	9	75,6
R16	Measurement, inspection, and verification errors	175	7	61,25
R17	Delays in approval processes	336	6	100,8
R18	Adverse weather conditions	90	10	45
R19	Employer-related issues	288	14	201,6
R20	Uncontrollable external factors	180	17	153

Recognizing that both RPN values and structural influence patterns may vary across different projects and risk sets, the proposed integrated approach was operationalized through the development of a decision support tool. This tool, named PREVIA, enables the application of the integrated RPN–MICMAC framework to different project contexts by allowing users to select project-specific risk factors and generate customized prioritization results.

5.4.3. Development of the PREVIA: Decision Making Support Tool for Risk Management

5.4.3.1. Operation of the PREVIA System

This section presents the development of PREVIA, risk-based decision support tool designed to facilitate project-specific risk analysis within an organizational context. The primary objective of the tool is to support decision-makers by integrating structural risk relationships and quantitative risk prioritization into a unified analytical framework. By combining ISM, MICMAC, and RPN methodologies within a single application, the proposed tool enables a systematic and comprehensive evaluation of risks tailored to the specific characteristics of individual projects.

Before detailing the user interaction and analytical components of the tool, the overall workflow is presented to provide a holistic view of the user–system interaction and the underlying analytical process. This visual representation helps clarify how user-defined inputs are transformed into structured analytical outputs through successive processing stages. The conceptual workflow of the proposed decision support tool is illustrated in Figure 6.

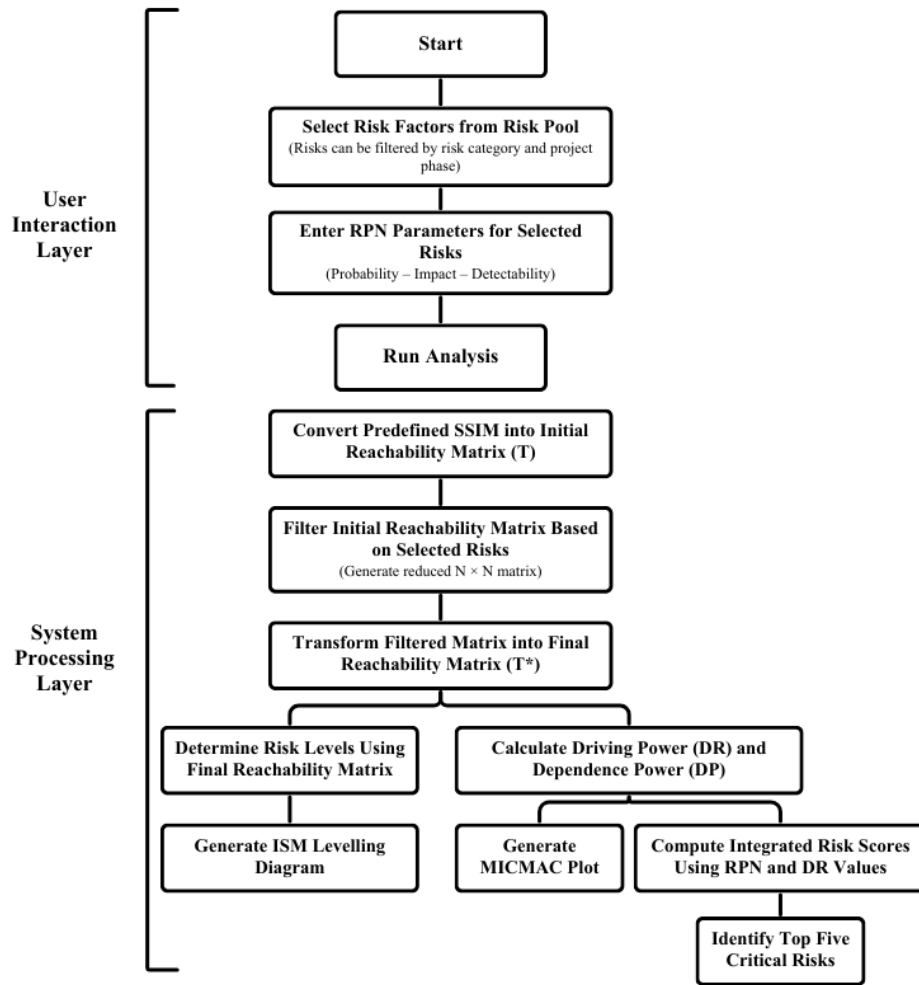
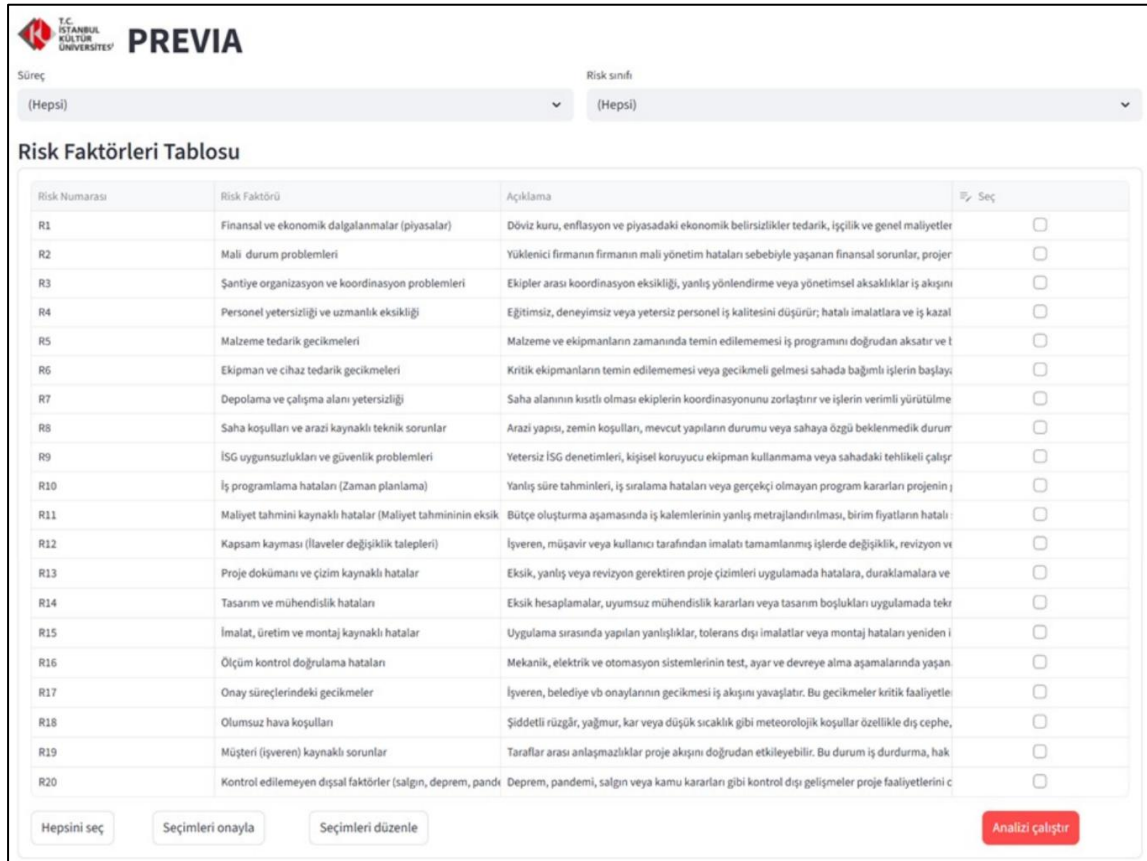


Figure 6. Workflow of the PREVIA risk-based decision support tool

The application is built upon a predefined pool of 20 risk factors, which are incorporated into the system as a demo dataset representing a reference risk repository for the organization. Rather than treating these risks as a fixed list, the tool allows users to dynamically interact with the risk pool by selecting only those risks that are relevant to their specific project conditions. The risk pool can be filtered based on risk category and the project phase to which each risk pertains, enabling a more focused and context-sensitive risk selection process. This flexibility ensures that the analysis reflects project-specific risk profiles rather than a generic assessment. The risk selection interface of the decision support tool is illustrated in Figure 7. Also Figure 8 shows the filtered table for selected options.

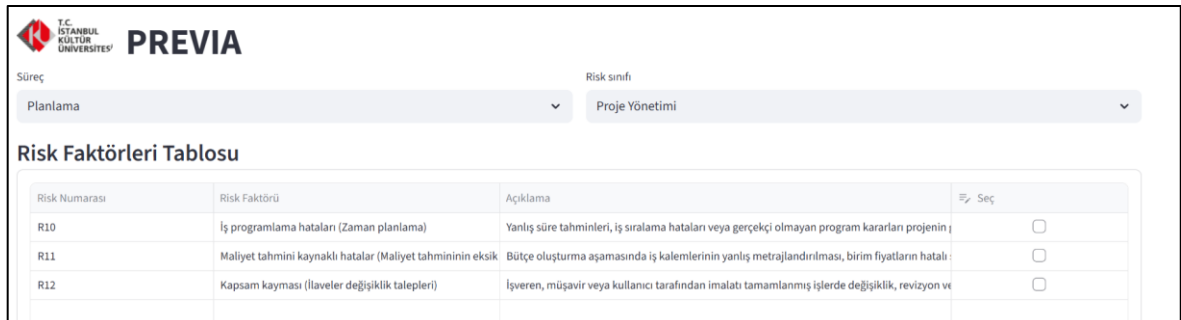


Risk Faktörleri Tablosu

Risk Numarası	Risk Faktörü	Açıklama	Seç
R1	Finansal ve ekonomik dalgalanmalar (piyasalar)	Döviz kuru, enflasyon ve piyasadaki ekonomik belirsizlikler tedarik, işçilik ve genel maliyetler	☐
R2	Mali durum problemleri	Yüklenici firmanın firmanın mali yönetim hataları sebebiyle yaşanan finansal sorunlar, projeler	☐
R3	Şantiye organizasyon ve koordinasyon problemleri	Ekipler arası koordinasyon eksikliği, yanlış yönlendirme veya yönetsel aksaklıklar iş akışını	☐
R4	Personel yetersizliği ve uzmanlık eksikliği	Eğitimsiz, deneyimsiz veya yetersiz personel iş kalitesini düşürür; hatalı imalatlar ve iş kazaları	☐
R5	Malzeme tedarik gecikmeleri	Malzeme ve ekipmanların zamanında temin edilememesi iş programını doğrudan aksatır ve t	☐
R6	Ekipman ve cihaz tedarik gecikmeleri	Kritik ekipmanların temin edilememesi veya gecikmeli gelmesi sahada bağımlı işlerin başla	☐
R7	Depolama ve çalışma alanı yetersizliği	Saha alanının kısıtlı olması ekiplerin koordinasyonunu zorlaştırır ve işlerin verimli yürütülme	☐
R8	Saha koşulları ve arazi kaynaklı teknik sorunlar	Arazi yapısı, zemin koşulları, mevcut yapıların durumu veya sahaya özgü beklenmedik durum	☐
R9	İSG uygunsuzlukları ve güvenlik problemleri	Yetersiz İSG denetimleri, kişisel koruyucu ekipman kullanmama veya sahadaki tehlikeli çalış	☐
R10	İş programlama hataları (Zaman planlama)	Yanlış süre tahminleri, iş sıralama hataları veya gerçekleşmeyen program kararları projenin	☐
R11	Maliyet tahmini kaynaklı hatalar (Maliyet tahmininin eksik	Bütçe oluşturma aşamasında iş kalemlerinin yanlış metrajlandırılması, birim fiyatların hatalı	☐
R12	Kapsam kayması (İlaveler değişikliği talepleri)	İşveren, müşavir veya kullanıcı tarafından imalatı tamamlanmış işlerde değişiklik, revizyon ve	☐
R13	Proje dokümanı ve çizim kaynaklı hatalar	Eksik, yanlış veya revizyon gerektiren proje çizimleri uygulamada hatalara, duraklamalara ve	☐
R14	Tasarım ve mühendislik hataları	Eksik hesaplamalar, uyumsuz mühendislik kararları veya tasarım boşlukları uygulamada tekr	☐
R15	İmalat, üretim ve montaj kaynaklı hatalar	Uygulama sırasında yapılan yanlışlıklar, tolerans dışı imalatlar veya montaj hataları yeniden i	☐
R16	Ölçüm kontrol doğrulama hataları	Mekanik, elektrik ve otomasyon sistemlerinin test, ayar ve devreye alma aşamalarında yaşan	☐
R17	Onay süreçlerindeki gecikmeler	İşveren, belediye vb onaylarının gecikmesi iş akışını yavaşlatır. Bu gecikmeler kritik faaliyetle	☐
R18	Olumsuz hava koşulları	Şiddetli rüzgâr, yağmur, kar veya düşük sıcaklık gibi meteorolojik koşullar özellikle dış cephe,	☐
R19	Müşteri (işveren) kaynaklı sorunlar	Taraflar arası anlaşmazlıklar proje akışını doğrudan etkileyebilir. Bu durum iş durdurma, hak	☐
R20	Kontrol edilemeyen dışsal faktörler (salgın, deprem, pandemi)	Deprem, pandemi, salgın veya kamu kararları gibi kontrol dışı gelişmeler proje faaliyetlerini c	☐

Hepsini seç Seçimleri onayla Seçimleri düzenle Analizi çalıştır

Figure 7. PREVIA interface view



Risk Faktörleri Tablosu

Risk Numarası	Risk Faktörü	Açıklama	Seç
R10	İş programlama hataları (Zaman planlama)	Yanlış süre tahminleri, iş sıralama hataları veya gerçekleşmeyen program kararları projenin	☐
R11	Maliyet tahmini kaynaklı hatalar (Maliyet tahmininin eksik	Bütçe oluşturma aşamasında iş kalemlerinin yanlış metrajlandırılması, birim fiyatların hatalı	☐
R12	Kapsam kayması (İlaveler değişikliği talepleri)	İşveren, müşavir veya kullanıcı tarafından imalatı tamamlanmış işlerde değişiklik, revizyon ve	☐

Hepsini seç Seçimleri onayla Seçimleri düzenle Analizi çalıştır

Figure 8. Table of filtered risk factors

Following the risk selection stage, the user interaction process continues with the input of RPN parameters for the selected risks. For each risk factor, the user assigns scores for probability of occurrence, impact severity, and detectability. This step allows expert judgment to be systematically incorporated into the analysis and ensures that project-specific perceptions are directly reflected in the quantitative evaluation. By structuring expert input within a standardized scoring framework, the tool supports consistent data entry while preserving the flexibility required to account for varying project conditions. The interface used for entering RPN parameters for the selected risks is shown in Figure 9.

T.C. İSTANBUL KÜLTÜR ÜNİVERSİTESİ' PREVIA

Süreç: (Hepsi) Risk sınıfı: (Hepsi)

Risk Faktörleri Tablosu

Risk Numarası	Risk Faktörü	Açıklama	Olasılık	Etki	Tespit
R3	Şantiye organizasyon ve koordinasyon problemleri	Ekipler arası koordinasyon eksikliği, yanlış yönlendirme veya yönetsel aksaklıklar	10	10	10
R4	Personel yetersizliği ve uzmanlık eksikliği	Eğitimsiz, deneyimsiz veya yetersiz personel iş kalitesini düşürür; hatalı imalatlar	10	10	10
R5	Malzeme tedarik gecikmeleri	Malzeme ve ekipmanların zamanında temin edilememesi iş programını doğrudan etkiler	10	10	10
R6	Ekipman ve cihaz tedarik gecikmeleri	Kritik ekipmanların temin edilememesi veya gecikmeli gelmesi sahada bağımlı işleri etkiler	10	10	10
R7	Depolama ve çalışma alanı yetersizliği	Saha alanının kısıtlı olması ekiplerin koordinasyonunu zorlaştırır ve işlerin verimliliğini düşürür	10	10	10
R8	Saha koşulları ve arazi kaynaklı teknik sorunlar	Arazi yapısı, zemin koşulları, mevcut yapıların durumu veya sahaya özgü beklenmeyen durumlar	10	10	10
R9	İSG uygunsuzlukları ve güvenlik problemleri	Yetersiz İSG denetimleri, kişisel koruyucu ekipman kullanmama veya sahadaki tarama eksiklikleri	10	10	10
R10	İş programlama hataları (Zaman planlama)	Yanlış süre tahminleri, iş sıralama hataları veya gerçekçi olmayan program kararları	10	10	10
R11	Maliyet tahmini kaynaklı hatalar (Maliyet tahmini)	Bütçe oluşturma aşamasında iş kalemlerinin yanlış metrajlandırılması, birim fiyat hataları	10	10	10
R12	Kapsam kayması (İlave değişiklik talepleri)	İşveren, müşavir veya kullanıcı tarafından imalatı tamamlanmış işlerde değişiklik talepleri	10	10	10
R12	Kapsam kayması (İlave değişiklik talepleri)	İşveren, müşavir veya kullanıcı tarafından imalatı tamamlanmış işlerde değişiklik talepleri	10	10	10
R13	Proje dokümanı ve çizim kaynaklı hatalar	Eksik, yanlış veya revizyon gerektiren proje çizimleri uygulamada hatalara, duraklamalara sebep olur	10	10	10
R14	Tasarım ve mühendislik hataları	Eksik hesaplamalar, uyumsuz mühendislik kararları veya tasarım boşlukları uygulamada sorunlara sebep olur	10	10	10
R15	İmalat, üretim ve montaj kaynaklı hatalar	Uygulama sırasında yapılan yanlışlıklar, tolerans dışı imalatlar veya montaj hataları	10	10	10
R16	Ölçüm kontrol doğrulama hataları	Mekanik, elektrik ve otomasyon sistemlerinin test, ayar ve devreye alma aşamalarında hatalar	10	10	10
R17	Onay süreçlerindeki gecikmeler	İşveren, belediye vb onaylarının gecikmesi iş akışını yavaşlatır. Bu gecikmeler krizlere sebep olabilir	10	10	10
R18	Olumsuz hava koşulları	Şiddetli rüzgâr, yağmur, kar veya düşük sıcaklık gibi meteorolojik koşullar özelliği	10	10	10
R19	Müşteri (işveren) kaynaklı sorunlar	Taraflar arası anlaşmazlıklar proje akışını doğrudan etkileyebilir. Bu durum iş duraklamalarına sebep olur	10	10	10
R20	Kontrol edilemeyen dışsal faktörler (salgın, deprem, pandemi)	Deprem, pandemi, salgın veya kamu kararları gibi kontrol dışı gelişmeler proje faaliyetlerini etkiler	10	10	10

Hepsini seç Seçimleri onayla Seçimleri düzenle Analizi çalıştır

Figure 9. RPN section of selected risk factors

Once the analysis is initiated, the system performs a series of predefined analytical steps in the background. The SSIM defined within the application is first converted into the initial reachability matrix. This matrix is then filtered according to the risks selected by the user, resulting in a reduced $N \times N$ matrix corresponding only to the project-specific risk set. The filtered matrix is subsequently transformed into the final reachability matrix, which serves as the common input for both ISM and MICMAC analyses. Using this matrix, the hierarchical levels of the risk factors are determined and the ISM levelling structure is generated. In parallel, driving and dependence power values are calculated to support the MICMAC analysis. In addition, the RPN values provided by the user are combined with MICMAC-based influence measures to derive integrated risk scores.

Upon completion of the analysis, the decision support tool generates three main outputs that collectively support risk-related decision-making. The first output is the ISM levelling diagram, which illustrates the hierarchical structure and interdependencies among the selected risk factors. The second output is the MICMAC plot, providing a visual representation of the influence–dependence characteristics of the risks. The third output is

an integrated risk prioritization table, in which RPN values and MICMAC-based influence measures are jointly evaluated to identify the five most critical risks for the project under consideration. These outputs enable users to simultaneously assess risk relationships, relative importance, and critical focus areas. The output interface displaying these results is shown in Figure 10.

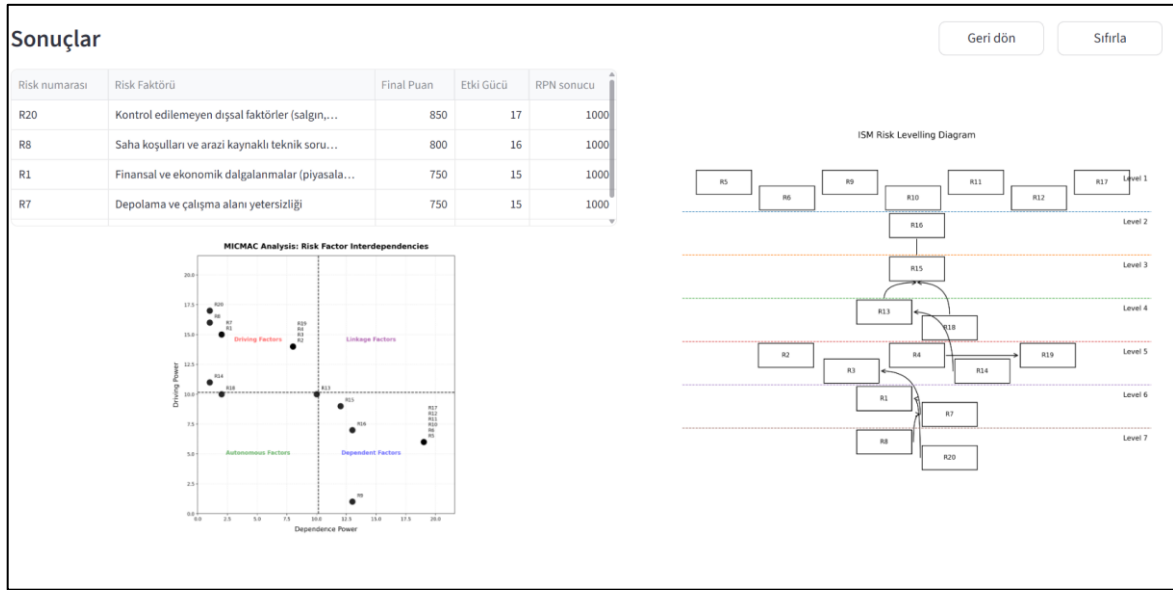


Figure 10. Visualization of risk analysis and interdependencies in PREVIA

From a decision support perspective, the proposed tool extends beyond conventional risk listing approaches by delivering prioritized and visually structured information. By presenting both relational and quantitative insights within a single analytical environment, the application assists project managers and senior decision-makers in identifying leverage risks, allocating resources more effectively, and focusing mitigation efforts on the most critical risk areas. The integrated nature of the outputs enhances interpretability and supports more informed and transparent decision-making processes.

In terms of organizational use, the decision support tool is designed to be reusable across different projects and adaptable to varying project conditions. The same risk pool can be employed to generate distinct analytical outcomes for different projects by modifying risk selection and RPN inputs. This capability enables scenario-based evaluations and supports the integration of the tool into broader corporate risk management practices. As such, the application demonstrates strong potential for institutionalization within organizational risk assessment workflows. Overall, the developed decision support tool represents the practical outcome of the proposed methodological framework and demonstrates its applicability in an

organizational setting. By translating the integrated ISM–MICMAC–RPN approach into an operational tool, this study bridges the gap between methodological development and practical implementation, providing a foundation for the discussion of results and implications in the subsequent sections.

5.4.3.2. Technical Implementation Notes

The decision support tool was implemented in Python using a modular architecture in which each analytical component (SSIM processing, reachability computation, ISM levelling, MICMAC analysis, and integrated prioritization) is encapsulated as a separate module. This design improves maintainability and supports extensibility, as individual methods can be updated or replaced without affecting the overall workflow. The main application script orchestrates the end-to-end pipeline by calling these dedicated modules sequentially, ensuring a clear separation between user interaction logic and analytical computations.

The interactive interface was developed using Streamlit, which enables rapid development of data-driven web applications and allows users to perform risk selection, parameter entry, and analysis execution within a single environment. User inputs and intermediate states are managed through Streamlit’s session management utilities, allowing the application to preserve selections and parameter values across pages and interactions. To improve responsiveness, frequently used resources such as the risk catalog and the SSIM-derived adjacency matrix are handled via caching mechanisms, reducing repeated file I/O and computation.

For data manipulation and validation, pandas is employed extensively to manage risk tables, filter matrices based on user-selected risks, and compute derived metrics such as RPN values. The plotting of analytical outputs is handled through Matplotlib (invoked within the corresponding modules), enabling the automated generation of the ISM levelling diagram and the MICMAC plot as image outputs. These figures are then rendered within the application interface. Additionally, the final prioritization output is produced as a structured table that highlights the top-ranked risks according to the integrated scoring approach.

Although the core implementation is largely procedural at the application level, the overall structure reflects an object-oriented design rationale through modular decomposition and clear responsibility allocation across modules. This approach supports consistent execution of the analytical workflow while keeping the user interface layer independent from the analytical logic.

6. CONCLUSION

This study is motivated by the problem that risks encountered in construction projects are often evaluated as isolated and independent events, which limits the understanding of their actual impacts in complex project environments. In particular, when risks are able to trigger one another and generate cascading effects, individual assessments based solely on probability and impact may become insufficient. Within this context, the study aimed to demonstrate the necessity of addressing risks by considering their interactions and cause–effect relationships within the overall project system. To support this perspective, the proposed analytical framework is operationalized through the development of a decision support tool, enabling the practical application of the integrated approach.

A total of 479 risk records from ten different projects are analyzed, and risks with similar characteristics are classified under 20 potential state categories. To identify cause–effect relationships among risks and to reveal their hierarchical structure, the Interpretive Structural Modeling (ISM) method is employed, while the impact and dependency levels of risks are evaluated using MICMAC (Matrice d'Impacts Croisés Multiplication Appliquée à un Classement) analysis. As a result, 20 risk factors are structured under 7 levels and grouped under 4 categories based on their dependence and driving powers. Furthermore, the classical Risk Priority Number (RPN) method is integrated into the proposed risk analysis model. A decision support system named PREVIA (Project Risk Evaluation and Visual Impact Analyzer) is also developed that allows a visual analysis of risks together with their hierarchical levels, mutual interactions, and priority rankings.

The findings of the study indicate that the roles of risks within a project system cannot be adequately defined only through numerical priority values. The analysis has shown that certain risks, although appearing to be of relatively low importance when evaluated individually, may assume a decisive role in the project due to their triggering effects on other risks. This observation suggests that risk prioritization structures can change significantly when risks are treated as interacting elements rather than isolated events. In particular, it has been observed that addressing driving risks at an early stage may lead to indirect improvements across multiple dependent risks. These findings are further reflected in the outputs generated through the developed decision support tool, which enabled the visualization and prioritization of such systemic risk effects in a project-oriented manner. These findings are generally consistent with previous studies that emphasize the relational

and system-based nature of risks. At the same time, evaluating the structural positions of risks together with quantitative prioritization has demonstrated that risk management can move beyond being a result-oriented ranking exercise toward a more analytical decision-support process that makes cause–effect relationships explicit. In this respect, the implementation of the proposed framework through the PREVIA decision support tool illustrates how such an analytical perspective can be translated into a usable and transparent decision-making environment for practitioners.

The main contribution of this study lies in proposing a systematic framework that integrates relational and quantitative risk assessment approaches for construction projects and in demonstrating its applicability through a dedicated decision support tool. The proposed approach enables decision-makers to focus not only on risks with high priority scores, but also on those that play a guiding role within the project system. By embedding this integrated logic into PREVIA, risk management processes can be positioned on a more proactive, consistent, and strategic basis. Accordingly, the study provides a methodological contribution to the academic literature while also offering a practical perspective to support risk-related decision-making.

Nevertheless, this study has certain limitations. The analyses were conducted using risk data specific to construction projects and limited to a particular time period. In addition, relationships among risks were examined within a static structure, and potential dynamic interactions that may evolve throughout the project lifecycle were not considered. The decision support tool developed within the scope of this study was implemented primarily for demonstration purposes and has not yet been validated through long-term organizational use. These limitations should be taken into account when interpreting the generalizability of the results.

Future studies may extend this research by testing the proposed approach and the PREVIA decision support tool with larger and more diverse datasets and by incorporating dynamic models that capture the temporal evolution of risk interactions. Furthermore, the framework can be enriched through advanced weighting techniques and alternative decision-support mechanisms. Expanding the application of the approach and the associated tool beyond construction projects to other project management domains—such as information technology, logistics, and service sectors—would further enhance its validity and applicability.

REFERENCES

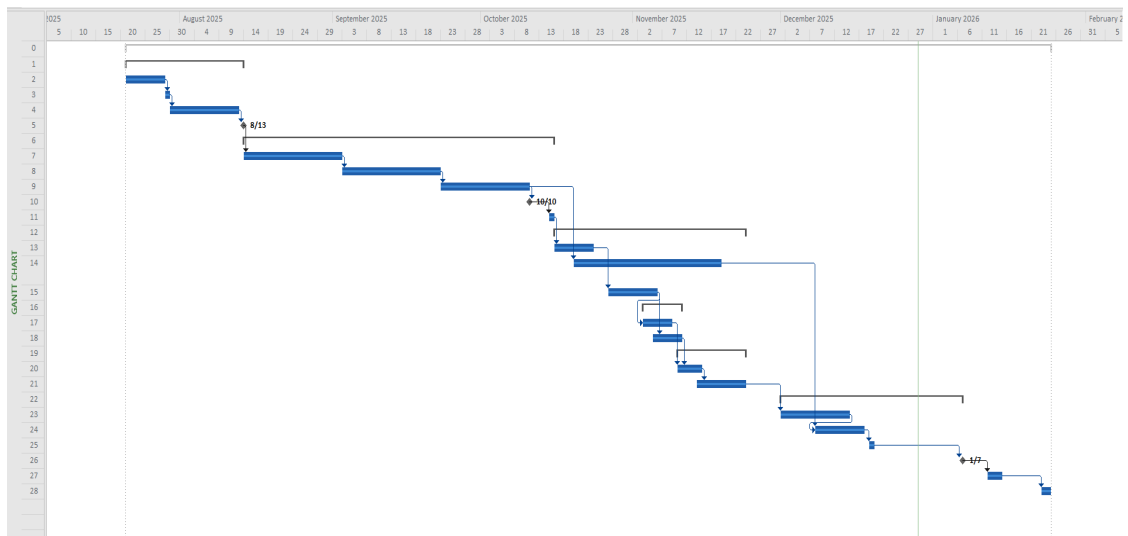
- Almasri, M. V. E. (2019). *The critical success factors of sustainable supply chain management practices and their effects on sustainability performance* (Doctoral dissertation). Akdeniz University, Graduate School of Social Sciences, Antalya, Turkey.
- Carvalho, M. M., & Rabechini Junior, R. (2015). Impact of risk management on project performance. *International Journal of Project Management*, 33(4), 791–803.
- Çakırlı, M. Y. (2017). *Structural modeling and analysis of barriers to ERP implementation* (Master's thesis). Istanbul Technical University, Graduate School of Science, Istanbul, Turkey.
- Çelebi, D. (2021). *Social entrepreneurship in gastronomy industry: A holistic approach* (Doctoral dissertation). Yaşar University, Graduate School, İzmir, Turkey.
- Eskerod, P., Jepsen, A. L., & Savage, G. (2015). Project stakeholder management—Past and present. *Project Management Journal*, 46(6), 6–19.
- Ergün, A. M. (2025). *A conceptual model based on SMART ISM–MICMAC analysis for lean and green (LG) management system implementation focusing on SMEs: Automotive industry application* (Master's thesis). Istanbul Technical University, Graduate School, Istanbul, Turkey.
- Guan, L., Abbasi, A., & Ryan, M. J. (2020). Analyzing green building project risk interdependencies using interpretive structural modeling. *Journal of Cleaner Production*, 256, 120372. doi:10.1016/j.jclepro.2020.120372
- Hansen, S., Mowen, M. M., & Heitger, D. L. (2021). *Cornerstones of managerial accounting* (7th ed.). Boston, MA: Cengage Learning.
- Heagney, J. (2016). *Fundamentals of Project Management* (4th ed.). American Management Association.
- Hillson, D., & Simon, P. (2020). *Practical project risk management: The ATOM methodology* (3rd ed.). Vienna, VA: Management Concepts.
- Karadayı, S. (2014). *Risk analysis in service supply chains* (Master's thesis). Istanbul Technical University, Graduate School of Science, Istanbul, Turkey.

- Kerzner, H. (2017). *Project management: A systems approach to planning, scheduling, and controlling* (12th ed.). Hoboken, NJ: John Wiley & Sons.
- Mansuroğlu, E. (2021). *Determination of competitive strategy and firm performance relationship in Ro-Ro companies operating in Turkey* (Master's thesis). Dokuz Eylul University, Graduate School of Social Sciences, İzmir, Turkey.
- Nikjow, M. A., Liang, L., Xijing, Q., & Sonar, H. (2022). Risk analysis of Belt and Road infrastructure projects using integrated ISM–MICMAC approach. *Journal of Modelling in Management*, 17(4), 1410–1431. doi:10.1108/JM2-02-2021-0038
- Project Management Institute. (2017). *A guide to the project management body of knowledge (PMBOK® Guide)* (6th ed.). Newtown Square, PA: Project Management Institute.
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® Guide)* (7th ed.). Newtown Square, PA: Project Management Institute.
- Sağ, S. (2022). *Open innovation management: Development of an open innovation index* (Doctoral dissertation). Gebze Technical University, Institute of Social Sciences, Gebze, Turkey.
- Thammasi, K. (2018). Project procurement management: Processes and challenges. *International Journal of Project Management*, 36(2), 287–299.
- Tuncer, Y. (2024). *Artificial intelligence based risk analysis in project management* (Master's thesis). Marmara University, Institute for Graduate Studies in Pure and Applied Sciences, İstanbul, Turkey.
- Ünlü, D. B. (2017). *Analysis of factors that affect transport mode choice decision in logistics* (Master's thesis). Akdeniz University, Graduate School of Social Sciences, Antalya, Turkey.
- Yayla, S. (2019). *Classification of factors that affect IT project success and determine root cause* (Master's thesis). Bahçeşehir University, Institute of Social Sciences, İstanbul, Turkey.

- Yıldırım, S. (2022). *Evaluation of green innovation adoption barriers of SMEs via fuzzy ISM–MICMAC–DEMATEL* (Master’s thesis). Bilecik Şeyh Edebali University, Institute of Social Sciences, Bilecik, Turkey.
- Yomralıoğlu, T. (2022). *Service quality indicators in airlines in Turkey: A weighted interpretive structural modeling approach* (Master’s thesis). Ondokuz Mayıs University, Institute of Graduate Studies, Samsun, Turkey.
- Zhang, G., Zou, P. X. W., & Wang, J. (2007). Identification and classification of construction risks in China. *Journal of Construction Engineering and Management*, 133(4), 328–336.
- Zou, P. X. W., Zhang, G., & Wang, J. (2007). Understanding the key risks in construction projects in China. *International Journal of Project Management*, 25(6), 601–614.
- Zulch, B. (2014). Communication: The foundation of project management. *Procedia Technology*, 16, 1000–1009.

APPENDIX A- PROJECT SCHEDULE

	Task Mode	Task Name	Baseline Start	Baseline Finish	Actual Start	Actual Finish	% Work	Duration	Start	Finish
0	✓	IE8900 (project schedule_plan)	Mon 7/21/25	Sat 1/31/26	Mon 7/21/25	Sat 1/24/26	100%	188 days	Mon 7/21/25	Sat 1/24/26
1	✓	1 Initiation Phase	Mon 7/21/25	Wed 8/13/25	Mon 7/21/25	Wed 8/13/25	100%	24 days	Mon 7/21/25	Wed 8/13/25
2	✓	1.1 Meetings with possible advisors	Mon 7/21/25	Mon 7/28/25	Mon 7/21/25	Mon 7/28/25	100%	8 days	Mon 7/21/25	Mon 7/28/25
3	✓	1.2 Submission of advisor preference form	Tue 7/29/25	Tue 7/29/25	Tue 7/29/25	Tue 7/29/25	100%	1 day	Tue 7/29/25	Tue 7/29/25
4	✓	1.3 Matching advisors&project teams	Wed 7/30/25	Tue 8/12/25	Wed 7/30/25	Tue 8/12/25	100%	14 days	Wed 7/30/25	Tue 8/12/25
5	✓	1.4 Submission of advisor agreement form	Wed 8/13/25	Wed 8/13/25	Wed 8/13/25	Wed 8/13/25	100%	0 days	Wed 8/13/25	Wed 8/13/25
6	✓	2 Proposal Phase	Thu 8/14/25	Wed 10/15/25	Thu 8/14/25	Wed 10/15/25	100%	63 days	Thu 8/14/25	Wed 10/15/25
7	✓	2.1 Searching company and meeting with the companies	Thu 8/14/25	Tue 9/2/25	Thu 8/14/25	Tue 9/2/25	100%	20 days	Thu 8/14/25	Tue 9/2/25
8	✓	2.2 Determining the graduation project company	Wed 9/3/25	Mon 9/22/25	Wed 9/3/25	Mon 9/22/25	100%	20 days	Wed 9/3/25	Mon 9/22/25
9	✓	2.3 Identifying project title, scope and methodology	Tue 9/23/25	Fri 10/10/25	Tue 9/23/25	Fri 10/10/25	100%	18 days	Tue 9/23/25	Fri 10/10/25
10	✓	2.4 Submission of proposal form	Fri 10/10/25	Fri 10/10/25	Fri 10/10/25	Fri 10/10/25	100%	0 days	Fri 10/10/25	Fri 10/10/25
11	✓	2.5 Proposal presentations	Wed 10/15/25	Wed 10/15/25	Wed 10/15/25	Wed 10/15/25	100%	1 day	Wed 10/15/25	Wed 10/15/25
12	✓	3 Progress Phase	Sat 10/11/25	Thu 11/23/25	Thu 10/16/25	Sun 11/23/25	100%	39 days	Thu 10/16/25	Sun 11/23/25
13	✓	3.1 Prepare the detailed scope (content)	Thu 10/16/25	Thu 10/23/25	Thu 10/16/25	Thu 10/23/25	100%	8 days	Thu 10/16/25	Thu 10/23/25
14	✓	3.2 Make Literature Review on Risk Management, ISM and MICMAC	Sat 10/11/25	Sun 11/9/25	Mon 10/20/25	Tue 11/18/25	100%	30 days	Mon 10/20/25	Tue 11/18/25
15	✓	3.3 Identify of Risks with Company Experts	Fri 10/24/25	Sun 11/2/25	Mon 10/27/25	Wed 11/5/25	100%	10 days	Mon 10/27/25	Wed 11/5/25
16	✓	3.4 Applying Risk Analysis Using the ISM Method	Mon 11/3/25	Sat 11/8/25	Mon 11/3/25	Mon 11/10/25	100%	8 days	Mon 11/3/25	Mon 11/10/25
17	✓	3.4.1 Develop a Structural Self-Interaction Matrix (SSIM)	Mon 11/3/25	Sat 11/8/25	Mon 11/3/25	Sat 11/8/25	100%	6 days	Mon 11/3/25	Sat 11/8/25
18	✓	3.4.2 Develop a Reachability Matrix (RM)	Mon 11/3/25	Sat 11/8/25	Wed 11/5/25	Mon 11/10/25	100%	6 days	Wed 11/5/25	Mon 11/10/25
19	✓	3.5 Applying MICMAC Analysis	Sun 11/9/25	Sun 11/23/25	Mon 11/10/25	Sun 11/23/25	100%	14 days	Mon 11/10/25	Sun 11/23/25
20	✓	3.5.1 Calculate the Drive-Dependence of Risk Interdependence	Sun 11/9/25	Thu 11/13/25	Mon 11/10/25	Fri 11/14/25	100%	5 days	Mon 11/10/25	Fri 11/14/25
21	✓	3.5.2 Draw a Drive-Dependence Diagram	Fri 11/14/25	Sun 11/23/25	Fri 11/14/25	Sun 11/23/25	100%	10 days	Fri 11/14/25	Sun 11/23/25
22	✓	4 Finalization Phase	Mon 11/24/25	Mon 12/1/25	Mon 12/1/25	Wed 1/7/26	100%	37 days	Mon 12/1/25	Wed 1/7/26
23	✓	4.1 Interpret the Results	Mon 11/24/25	Sun 12/7/25	Mon 12/1/25	Sun 12/14/25	100%	14 days	Mon 12/1/25	Sun 12/14/25
24	✓	4.2 Finish the report and writing conclusion	Mon 12/8/25	Wed 12/17/25	Mon 12/8/25	Wed 12/17/25	100%	10 days	Mon 12/8/25	Wed 12/17/25
25	✓	4.3 Submission of final report	Fri 12/19/25	Fri 12/19/25	Fri 12/19/25	Fri 12/19/25	100%	1 day	Fri 12/19/25	Fri 12/19/25
26	✓	4.4 Final Presentations	Mon 1/5/26	Mon 1/5/26	Wed 1/7/26	Wed 1/7/26	100%	0 days	Wed 1/7/26	Wed 1/7/26
27	✓	5 Revisions (If required)	Tue 1/6/26	Fri 1/30/26	Mon 1/12/26	Wed 1/14/26	100%	3 days	Mon 1/12/26	Wed 1/14/26
28	✓	6 Submission of 2 copies	Sat 1/31/26	Sat 1/31/26	Fri 1/23/26	Sat 1/24/26	100%	2 days	Fri 1/23/26	Sat 1/24/26



APPENDIX B - PREVIA SYSTEM SCRIPTS

```
# modules/ism_levels.py

from __future__ import annotations
import pandas as pd

def compute_ism_levels(reach_df: pd.DataFrame, *, include_self: bool = True) ->
list[list[str]]:
    if list(reach_df.index) != list(reach_df.columns):
        raise ValueError("reach_df kare olmalı (index/columns aynı olmalı).")

    remaining = list(reach_df.index)
    R = reach_df.astype(int).copy()

    levels: list[list[str]] = []

    while remaining:
        current_level: list[str] = []

        for i in remaining:
            reach_set = set(R.columns[R.loc[i] == 1])
            ant_set = set(R.index[R[i] == 1])

            if include_self:
                reach_set.add(i)
                ant_set.add(i)

            inter = reach_set & ant_set

            if reach_set == inter:
                current_level.append(i)

        if not current_level:
```

```
        raise RuntimeError("Seviye bulunamadı. Reachability/etiketlerde tutarsızlık  
olabilir.")
```

```
        levels.append(current_level)
```

```
        remaining = [x for x in remaining if x not in current_level]
```

```
        R = R.loc[remaining, remaining]
```

```
    return levels
```

```
# modules/ism_matplotlib_diagram.py
```

```
from __future__ import annotations
```

```
import os
```

```
import math
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
from matplotlib.patches import Rectangle, FancyArrowPatch
```

```
from typing import Literal
```

```
def _build_skeletal_edges(
```

```
    levels: list[list[str]],
```

```
    reach_df: pd.DataFrame,
```

```
    adj_df: pd.DataFrame
```

```
) -> list[tuple[str, str, str]]:
```

```

if reach_df.shape != adj_df.shape or list(reach_df.index) != list(adj_df.index):
    raise ValueError("reach_df ve adj_df aynı boyutta ve etiketlerde olmalıdır.")

```

```

N = len(reach_df)

```

```

R = reach_df.astype(int).to_numpy()

```

```

A = adj_df.astype(int).to_numpy()

```

```

nodes = list(reach_df.index)

```

```

S = A.copy()

```

```

for i in range(N):

```

```

    for j in range(N):

```

```

        if S[i, j] == 1 and i != j:

```

```

            for k in range(N):

```

```

                if k != i and k != j and A[i, k] == 1 and R[k, j] == 1:

```

```

                    S[i, j] = 0

```

```

                    break

```

```

np.fill_diagonal(S, 0)

```

```

directed: set[tuple[str, str]] = set()

```

```

node_to_level = {}

```

```

for li, nodes_in_level in enumerate(levels, start=1):

```

```

    for node in nodes_in_level:

```

```

        node_to_level[node] = li

```

```

for i in range(N):
    u = nodes[i]

    level_u = node_to_level.get(u)

    if level_u is None:
        continue

    for j in range(N):
        v = nodes[j]

        if S[i, j] == 1:
            level_v = node_to_level.get(v)

            if level_v is None:
                continue

            if level_v == level_u or level_v == level_u - 1:
                directed.add((u, v))

merged: list[tuple[str, str, str]] = []
seen: set[tuple[str, str]] = set()

for (u, v) in sorted(directed):
    if (u, v) in seen:
        continue

    if (v, u) in directed:
        a, b = sorted([u, v])

        merged.append((a, b, "both"))

        seen.add((u, v))

        seen.add((v, u))

```

```

else:

    merged.append((u, v, "one"))

    seen.add((u, v))

return merged


def _compute_layout(
    levels: list[list[str]],
    *,
    page_width: float = 14.0,
    page_height: float = 10.0,
    margin_x: float = 0.6,
    margin_y: float = 0.6,
    node_w: float = 1.6,
    node_h: float = 0.7,
) -> tuple[dict[str, tuple[float, float]], dict[int, tuple[float, float]]]:

    L = len(levels)

    if L == 0:

        return {}, {}

    usable_h = page_height - 2 * margin_y

    band_h = usable_h / L

    band_y: dict[int, tuple[float, float]] = {}

```

```

pos: dict[str, tuple[float, float]] = {}

usable_w = page_width - 2 * margin_x

stagger_offset = 0.5 * band_h - 0.5 * node_h - 0.05

min_gap_x = 0.3

for li in range(1, L + 1):

    top = page_height - margin_y - (li - 1) * band_h

    bottom = top - band_h

    band_y[li] = (bottom, top)

    nodes = levels[li - 1]

    k = len(nodes)

    if k == 0:

        continue

    y_center = (bottom + top) / 2.0

    required_w = k * node_w + (k - 1) * min_gap_x

    if required_w > usable_w:

        step = usable_w / k

        start_x = margin_x

        xs = [start_x + node_w / 2 + i * step for i in range(k)]

    else:

        start_x = margin_x + (usable_w - required_w) / 2.0

        xs = []

```

```

    current_x = start_x

    for _ in range(k):

        xs.append(current_x + node_w / 2.0)

        current_x += node_w + min_gap_x


    for idx, (n, x) in enumerate(zip(nodes, xs)):

        final_offset = max(0.0, stagger_offset)

        y = y_center + final_offset if idx % 2 == 0 else y_center - final_offset

        pos[n] = (x, y)


    return pos, band_y


def _anchor(
    pos: dict[str, tuple[float, float]],
    node_w: float,
    node_h: float,
    src: str,
    dst: str
) -> tuple[tuple[float, float], tuple[float, float]]:

    sx, sy = pos[src]

    dx, dy = pos[dst]

    padding = 0.05

    angle = math.atan2(dy - sy, dx - sx)

```

```

angle_deg = math.degrees(angle)

if -45 < angle_deg <= 45:
    p1 = (sx + node_w / 2.0 + padding, sy)
elif 45 < angle_deg <= 135:
    p1 = (sx, sy + node_h / 2.0 + padding)
elif angle_deg > 135 or angle_deg <= -135:
    p1 = (sx - node_w / 2.0 - padding, sy)
else:
    p1 = (sx, sy - node_h / 2.0 - padding)

angle_rev = math.atan2(sy - dy, sx - dx)
angle_rev_deg = math.degrees(angle_rev)

if -45 < angle_rev_deg <= 45:
    p2 = (dx + node_w / 2.0 + padding, dy)
elif 45 < angle_rev_deg <= 135:
    p2 = (dx, dy + node_h / 2.0 + padding)
elif angle_rev_deg > 135 or angle_rev_deg <= -135:
    p2 = (dx - node_w / 2.0 - padding, dy)
else:
    p2 = (dx, dy - node_h / 2.0 - padding)

return p1, p2

```

```

def draw_ism_level_diagram(
    levels: list[list[str]],
    reach_df: pd.DataFrame,
    adj_df: pd.DataFrame,
    out_path: str = "outputs/ism_level_diagram.png",
    *,
    title: str = "ISM Risk Levelling Diagram",
    page_width: float = 14.0,
    page_height: float = 10.0,
    node_w: float = 1.6,
    node_h: float = 0.7,
) -> str:

    if list(adj_df.index) != list(adj_df.columns):
        raise ValueError("adj_df kare olmalı (index==columns).")

    pos, band_y = _compute_layout(
        levels,
        page_width=page_width,
        page_height=page_height,
        node_w=node_w,
        node_h=node_h
    )

    edges = _build_skeletal_edges(levels, reach_df, adj_df)

```

```

fig, ax = plt.subplots(figsize=(page_width, page_height))

ax.set_xlim(0, page_width)

ax.set_ylim(0, page_height)

ax.axis("off")

ax.set_title(title, fontsize=14, pad=16)

L = len(levels)

for li in range(1, L):

    yb, yt = band_y[li]

    nb, nt = band_y[li + 1]

    y_line = (yb + nt) / 2.0

    ax.plot([0.2, page_width - 0.2], [y_line, y_line], linestyle="--", linewidth=1)

for li in range(1, L + 1):

    _, yt = band_y[li]

    ax.text(page_width - 0.3, yt - 0.2, f"Level {li}", ha="right", va="top", fontsize=11)

for n, (x, y) in pos.items():

    blx = x - node_w / 2.0

    bly = y - node_h / 2.0

    ax.add_patch(Rectangle((blx, bly), node_w, node_h, fill=False, linewidth=1.2))

    ax.text(x, y, n, ha="center", va="center", fontsize=10)

for u, v, kind in edges:

    if u not in pos or v not in pos:

```

```
continue
```

```
p1, p2 = _anchor(pos, node_w, node_h, u, v)
arrow_style = "<->" if kind == "both" else "->"
```

```
ax.add_patch(
    FancyArrowPatch(
        p1, p2,
        arrowstyle=arrow_style,
        mutation_scale=15,
        linewidth=1.2,
        connectionstyle="angle3",
        shrinkA=0,
        shrinkB=0
    )
)
```

```
os.makedirs(os.path.dirname(out_path) or ".", exist_ok=True)
fig.savefig(out_path, dpi=300, bbox_inches="tight")
plt.close(fig)
```

```
return out_path
```

```
# modules/matrix_filter.py

from __future__ import annotations

import pandas as pd
```

```

def filter_square_matrix(
    df: pd.DataFrame,
    selected: list[str],
    *,
    strict: bool = True,
    keep_order: bool = True,
) -> pd.DataFrame:
    if not isinstance(df, pd.DataFrame):
        raise TypeError("df pandas DataFrame olmalı.")
    if list(df.index) != list(df.columns):
        raise ValueError("Matrix kare olmalı ve index/columns aynı etiket/sırada olmalı.")
    if not selected:
        raise ValueError("selected boş olamaz. En az 1 eleman seçilmeli.")
    missing = [x for x in selected if x not in df.index]
    if missing and strict:
        raise ValueError(f"Seçilenlerden matriste olmayanlar var: {missing}")
    selected_in = [x for x in selected if x in df.index]
    if keep_order:
        return df.loc[selected_in, selected_in].copy()
    selected_set = set(selected_in)
    ordered = [x for x in df.index if x in selected_set]
    return df.loc[ordered, ordered].copy()

```

```

def assert_square_binary(df: pd.DataFrame) -> None:

```

```

if list(df.index) != list(df.columns):

    raise ValueError("Matrix kare değil (index/columns uyuşmuyor).")

vals = pd.unique(df.to_numpy().ravel())

vals = set(int(v) for v in vals if pd.notna(v))

if not vals.issubset({0, 1}):

    raise ValueError(f"Matrix binary değil. Bulunan değerler: {sorted(vals)}")

```

```

# modules/micmac.py

from __future__ import annotations


import os

from typing import Literal


import matplotlib.pyplot as plt

import numpy as np

import pandas as pd


QUADRANT_NAMES = {

    "driving": "Driving Factors",

    "linkage": "Linkage Factors",

    "dependent": "Dependent Factors",

    "autonomous": "Autonomous Factors",

}


def compute_micmac_analysis(reach_df: pd.DataFrame) -> pd.DataFrame:

    if list(reach_df.index) != list(reach_df.columns):

```

```

    raise ValueError("reach_df kare olmalı (index/columns aynı olmalı).")

driving_power = reach_df.sum(axis=1).rename("Driving_Power")
dependence_power = reach_df.sum(axis=0).rename("Dependence_Power")
micmac_df = pd.concat([driving_power, dependence_power], axis=1)
mean_dp = micmac_df["Driving_Power"].mean()
mean_dep = micmac_df["Dependence_Power"].mean()

def assign_quadrant(row: pd.Series) -> Literal["driving", "linkage", "dependent",
"autonomous"]:
    dp = row["Driving_Power"]
    dep = row["Dependence_Power"]
    if dp >= mean_dp and dep >= mean_dep:
        return "linkage"
    if dp >= mean_dp and dep < mean_dep:
        return "driving"
    if dp < mean_dp and dep >= mean_dep:
        return "dependent"
    return "autonomous"

micmac_df["Quadrant"] = micmac_df.apply(assign_quadrant, axis=1)
micmac_df["Mean_DP"] = mean_dp
micmac_df["Mean_DEP"] = mean_dep
return micmac_df

def draw_micmac_plot(

```

```

micmac_df: pd.DataFrame,

out_path: str = "outputs/micmac_plot.png",

*,

title: str = "MICMAC Analysis: Risk Factor Interdependencies",

figsize: tuple[int, int] = (10, 10),

) -> str:

    mean_dp = micmac_df["Mean_DP"].iloc[0]

    mean_dep = micmac_df["Mean_DEP"].iloc[0]

    fig, ax = plt.subplots(figsize=figsize)

    ax.scatter(

        micmac_df["Dependence_Power"],

        micmac_df["Driving_Power"],

        color="black",

        alpha=0.85,

        edgecolors="white",

        s=200,

    )

    grouped = micmac_df.groupby(["Dependence_Power", "Driving_Power"])

    base_dx, base_dy = 10, 10

    line_step = 12

    def _risk_sort_key(risk_name: str):

        try:

            if risk_name.startswith("R"):

                return (0, int(risk_name[1:]))

```

```

except Exception:

    pass

return (1, risk_name)

for (dep, dp), idxs in grouped.groups.items():

    names = sorted(list(idxs), key=_risk_sort_key)

    if len(names) == 1:

        i = names[0]

        ax.annotate(

            i,

            (dep, dp),

            xytext=(base_dx, base_dy),

            textcoords="offset points",

            fontsize=10,

            color="black",

        )

    else:

        for k, i in enumerate(names):

            ax.annotate(

                i,

                (dep, dp),

                xytext=(base_dx, base_dy + k * line_step),

                textcoords="offset points",

                fontsize=10,

                color="black",

            )

```

```

ax.axvline(mean_dep, color="k", linestyle="--", linewidth=1.5)
ax.axhline(mean_dp, color="k", linestyle="--", linewidth=1.5)
ax.set_xlabel("Dependence Power", fontsize=14, labelpad=10)
ax.set_ylabel("Driving Power", fontsize=14, labelpad=10)
ax.set_title(title, pad=15, fontsize=16, weight="bold")
ax.grid(True, linestyle=":", alpha=0.5)

max_val = micmac_df[["Driving_Power", "Dependence_Power"]].max().max()
pad_top = max(1.0, 0.08 * (max_val + 1))
ax.set_xlim(0, max_val + 1 + pad_top)
ax.set_ylim(0, max_val + 1 + pad_top)
ax.set_aspect("equal", adjustable="box")

colors = {
    "driving": "red",
    "linkage": "purple",
    "dependent": "blue",
    "autonomous": "green",
}

ax.text(
    mean_dep / 2,
    (mean_dp + max_val) / 2,
    QUADRANT_NAMES["driving"],
    ha="center",

```

```

        va="center",

        fontsize=12,

        weight="bold",

        color=colors["driving"],

        alpha=0.6,

    )

    ax.text(

        (mean_dep + max_val) / 2,

        (mean_dp + max_val) / 2,

        QUADRANT_NAMES["linkage"],

        ha="center",

        va="center",

        fontsize=12,

        weight="bold",

        color=colors["linkage"],

        alpha=0.6,

    )

    ax.text(

        (mean_dep + max_val) / 2,

        mean_dp / 2,

        QUADRANT_NAMES["dependent"],

        ha="center",

        va="center",

        fontsize=12,

        weight="bold",

        color=colors["dependent"],

```

```

        alpha=0.6,
    )
    ax.text(
        mean_dep / 2,
        mean_dp / 2,
        QUADRANT_NAMES["autonomous"],
        ha="center",
        va="center",
        fontsize=12,
        weight="bold",
        color=colors["autonomous"],
        alpha=0.6,
    )

    os.makedirs(os.path.dirname(out_path) or ".", exist_ok=True)

    fig.savefig(out_path, dpi=200, bbox_inches="tight")

    plt.close(fig)

    return out_path

```

```

# modules/reachability_iterative.py

from __future__ import annotations

import numpy as np
import pandas as pd

def boolean_matmul(X: np.ndarray, Y: np.ndarray) -> np.ndarray:

```

```

Xb = X.astype(bool, copy=False)

Yb = Y.astype(bool, copy=False)

Zb = (Xb[:, :, None] & Yb[None, :, :]).any(axis=1)

return Zb.astype(int)

```

```

def reachability_by_iterative_powers(binary_df: pd.DataFrame, *, max_iter: int | None =
None) -> pd.DataFrame:

    if list(binary_df.index) != list(binary_df.columns):

        raise ValueError("Matrix kare olmalı: index ve columns aynı etiket/sırada olmalı.")

    A = binary_df.astype(int).to_numpy()

    n = A.shape[0]

    if n == 0:

        return binary_df.copy()

    if n == 1:

        return binary_df.astype(int).copy()

    if max_iter is None:

        max_iter = max(1, n * n)

    cur = boolean_matmul(A, A)

    if np.array_equal(cur, A):

        return pd.DataFrame(cur, index=binary_df.index, columns=binary_df.columns)

    for _ in range(max_iter - 1):

        nxt = boolean_matmul(cur, A)

        if np.array_equal(nxt, cur):

            return pd.DataFrame(nxt, index=binary_df.index, columns=binary_df.columns)

        cur = nxt

```

```

    return pd.DataFrame(cur, index=binary_df.index, columns=binary_df.columns)

```

```

# modules/rpn_power_analysis.py

from __future__ import annotations

from typing import Dict, Union

import pandas as pd

def compute_rpn_weighted_power(
    micmac_df: pd.DataFrame,
    rpn_values: Union[Dict[str, int], pd.Series],
) -> pd.DataFrame:
    if isinstance(rpn_values, dict):
        rpn_series = pd.Series(rpn_values, name="RPN")
    else:
        rpn_series = rpn_values.rename("RPN")
    micmac_df = micmac_df.join(rpn_series, how="inner")
    if "Driving_Power" not in micmac_df.columns:
        raise ValueError("micmac_df Driving_Power sütununu içermelidir.")
    if "RPN" not in micmac_df.columns:
        raise ValueError("RPN değerleri yüklenemedi. Risk etiketlerini kontrol edin.")
    N = len(micmac_df)
    rpn_norm = micmac_df["RPN"] / 1000.0
    dp_norm = micmac_df["Driving_Power"] / N
    micmac_df["Weighted_Driving_Power"] = rpn_norm * dp_norm * 1000
    return micmac_df.sort_values(by="Weighted_Driving_Power", ascending=False)

```

```

# modules/ssim_loader.py

```

```

import pandas as pd

def load_ssim_csv(path: str) -> pd.DataFrame:
    df = pd.read_csv(path, sep=";", index_col=0)

    df.columns = [str(c).strip() for c in df.columns]
    df.index = [str(i).strip() for i in df.index]

    return df

```

```
# modules/ssim_to_binary.py
```

```
import pandas as pd
```

```
def convert_ssim_to_binary(df: pd.DataFrame) -> pd.DataFrame:
```

```
    risks = list(df.index)
```

```
    binary = pd.DataFrame(0, index=risks, columns=risks)
```

```
    for i in risks:
```

```
        for j in risks:
```

```
            if i == j:
```

```
                binary.loc[i, j] = 1
```

```
                continue
```

```
            if j not in df.columns:
```

```
                continue
```

```
            val = df.loc[i, j]
```

```
            if pd.isna(val):
```

```
                continue
```

```
            val = str(val).strip().upper()
```

```
            if val == "V":
```

```

        binary.loc[i, j] = 1

        binary.loc[j, i] = 0

    elif val == "A":

        binary.loc[i, j] = 0

        binary.loc[j, i] = 1

    elif val == "X":

        binary.loc[i, j] = 1

        binary.loc[j, i] = 1

    elif val == "O":

        binary.loc[i, j] = 0

        binary.loc[j, i] = 0

    return binary

```

```

# modules/risk_catalog.py

from __future__ import annotations

from pathlib import Path

import pandas as pd

def load_risk_catalog(
    xlsx_path: str | Path,
    sheet_name: str | int = 0,
) -> pd.DataFrame:

    xlsx_path = Path(xlsx_path)

    if not xlsx_path.exists():

        raise FileNotFoundError(f"Risk kataloĝu dosyası bulunamadı: {xlsx_path}")

    df = pd.read_excel(xlsx_path, sheet_name=sheet_name)

    df = df.loc[:, ~df.columns.astype(str).str.startswith("Unnamed")].copy()

```

```

df.columns = [str(c).strip() for c in df.columns]

df = df.rename(
    columns={
        "#": "No",
        "Süreç": "Surec",
        "Risk sınıfı": "Risk_Sinifi",
        "Risk Faktörü": "Risk_Adi",
        "Açıklama": "Aciklama",
    }
)

required = ["No", "Surec", "Risk_Sinifi", "Risk_Adi", "Aciklama"]
missing = [c for c in required if c not in df.columns]

if missing:
    raise ValueError(
        f'Risk kataloğunda eksik kolon(lar): {missing}. Mevcut: {list(df.columns)}'
    )

df["Risk_Adi"] = df["Risk_Adi"].astype(str).str.strip()

df["No"] = pd.to_numeric(df["No"], errors="raise").astype(int)

df["Risk_ID"] = df["No"].apply(lambda x: f'R{x}')

out = df[["Risk_ID", "No", "Surec", "Risk_Sinifi", "Risk_Adi", "Aciklama"]].copy()

return out

```

```

from __future__ import annotations

```

```

import base64

```

```

from pathlib import Path

```

```

import pandas as pd

import streamlit as st

from modules.ism_levels import compute_ism_levels

from modules.ism_matplotlib_diagram import draw_ism_level_diagram

from modules.matrix_filter import filter_square_matrix

from modules.micmac import compute_micmac_analysis, draw_micmac_plot

from modules.reachability_iterative import reachability_by_iterative_powers

from modules.rpn_power_analysis import compute_rpn_weighted_power

from modules.ssim_loader import load_ssim_csv

from modules.ssim_to_binary import convert_ssim_to_binary

from risk_catalog import load_risk_catalog


st.set_page_config(
    page_title="PREVIA",
    layout="wide",
    initial_sidebar_state="collapsed",
)


st.markdown(
    """
<style>

    .block-container { max-width: 99vw !important; padding-top: 0.25rem; padding-bottom:
0.25rem; }

    header[data-testid="stHeader"] { display: none; }

    footer { display: none; }

```

```

html, body, [data-testid="stAppViewContainer"], [data-testid="stApp"] {
  height: 100%;
  overflow: hidden !important;
}

[data-testid="stDataEditor"] { width: 100% !important; }

h1, h2, h3 { margin: 0.2rem 0 !important; padding: 0 !important; }

div[data-testid="stVerticalBlock"] > div { gap: 0.35rem; }

</style>

"",

unsafe_allow_html=True,

)

```

```

BASE_DIR = Path(__file__).resolve().parent
DATA_DIR = BASE_DIR / "Data"
OUT_DIR = BASE_DIR / "outputs"
SSIM_CSV = DATA_DIR / "ssimdata.csv"
RISK_XLSX = DATA_DIR / "risktablosu.xlsx"

```

```

def _ensure_session_defaults():
    st.session_state.setdefault("page", "input")
    st.session_state.setdefault("results", None)
    st.session_state.setdefault("ui_table", pd.DataFrame())
    st.session_state.setdefault("selection_locked", False)

```

```
@st.cache_data(show_spinner=False)

def _load_catalog() -> pd.DataFrame:

    return load_risk_catalog(RISK_XLSX)
```

```
@st.cache_data(show_spinner=False)

def _load_binary_adjacency() -> pd.DataFrame:

    ssim = load_ssim_csv(str(SSIM_CSV))

    adj = convert_ssim_to_binary(ssim)

    adj = adj.reindex(index=adj.columns, columns=adj.columns)

    return adj
```

```
@st.cache_data(show_spinner=False)

def _img_base64(path: str) -> str:

    p = Path(path)

    if not p.exists():

        return ""

    return base64.b64encode(p.read_bytes()).decode("utf-8")
```

```
def _img_html(path: str, *, max_h_vh: int) -> None:

    b64 = _img_base64(path)

    if not b64:

        st.info("Görsel çıktısı bulunamadı.")

    return
```

```

st.markdown(
    f"""
        <div style="width:100%; height:{max_h_vh}vh; display:flex; align-items:center;
justify-content:center;">

        </div>
    """,
    unsafe_allow_html=True,
)

```

```

def _init_ui_table_from_catalog(df: pd.DataFrame) -> pd.DataFrame:

```

```

    ui = df[["Risk_ID", "No", "Risk_Adi", "Aciklama"]].copy()

    ui["Seç"] = False

    ui["Olasilik"] = 10

    ui["Etki"] = 10

    ui["Tespit"] = 10

    return ui.sort_values("No").reset_index(drop=True)

```

```

def _merge_edited_into_full(full: pd.DataFrame, edited: pd.DataFrame) -> pd.DataFrame:

```

```

    if full.empty or "Risk_ID" not in edited.columns:

```

```

    return full

full_idx = full.set_index("Risk_ID")

edited_idx = edited.set_index("Risk_ID")

for col in edited.columns:

    if col != "Risk_ID" and col in full_idx.columns:

        full_idx.loc[edited_idx.index, col] = edited_idx[col]

return full_idx.reset_index()

```

```

def _compute_pipeline(selected_ids: list[str], rpn_df: pd.DataFrame) -> dict:

    OUT_DIR.mkdir(parents=True, exist_ok=True)

    adjacency = _load_binary_adjacency()

    A_sel = filter_square_matrix(adjacency, selected_ids)

    R = reachability_by_iterative_powers(A_sel)

    levels = compute_ism_levels(R)

    ism_path = str(OUT_DIR / "ism_level_diagram.png")

    draw_ism_level_diagram(

        levels=levels,

        reach_df=R,

        adj_df=A_sel,

        out_path=ism_path,

        title="ISM Risk Levelling Diagram",

    )

    mic_table = compute_micmac_analysis(R)

    micmac_plot_path = str(OUT_DIR / "micmac_plot.png")

    draw_micmac_plot(mic_table, out_path=micmac_plot_path)

```

```

rpn_df = rpn_df.copy()

rpn_df["Olasilik"] = pd.to_numeric(rpn_df["Olasilik"],
errors="coerce").fillna(1).astype(int)

rpn_df["Etki"] = pd.to_numeric(rpn_df["Etki"], errors="coerce").fillna(1).astype(int)
rpn_df["Tespit"] = pd.to_numeric(rpn_df["Tespit"], errors="coerce").fillna(1).astype(int)
rpn_df["RPN"] = rpn_df["Olasilik"] * rpn_df["Etki"] * rpn_df["Tespit"]
rpn_series = rpn_df.set_index("Risk_ID")["RPN"]
weighted = compute_rpn_weighted_power(mic_table, rpn_series).sort_values(
    "Weighted_Driving_Power", ascending=False
)
return {
    "selected_ids": selected_ids,
    "ism_plot_path": ism_path,
    "micmac_plot_path": micmac_plot_path,
    "weighted_table": weighted,
}

```

```

def page_input():

    logo_path = BASE_DIR / "assets" / "okullogo.png"

    col_logo, col_title = st.columns([1, 10], vertical_alignment="center")

    with col_logo:

        if logo_path.exists():

            st.image(str(logo_path), width=125)

    with col_title:

        st.title("PREVIA")

```

```
ALL_SURECLER = ["Başlatma", "Planlama", "Uygulama", "İzleme", "Kontrol",
"Kapanış"]
```

```
if not RISK_XLSX.exists():
```

```
    st.error(f"Risk Excel bulunamadı: {RISK_XLSX}")
```

```
    st.stop()
```

```
if not SSIM_CSV.exists():
```

```
    st.error(f"SSIM CSV bulunamadı: {SSIM_CSV}")
```

```
    st.stop()
```

```
catalog = _load_catalog()
```

```
f1, f2 = st.columns([1, 1])
```

```
with f1:
```

```
    surec = st.selectbox("Süreç", ["(Hepsi)"] + ALL_SURECLER)
```

```
with f2:
```

```
    sinif = st.selectbox(
```

```
        "Risk sınıfı",
```

```
        ["(Hepsi)"] + sorted(catalog["Risk_Sinifi"].dropna().unique().tolist()),
```

```
    )
```

```
import re
```

```
def _has_token(cell: str, token: str) -> bool:
```

```
    if cell is None or (isinstance(cell, float) and pd.isna(cell)):
```

```
        return False
```

```
    parts = [p.strip() for p in re.split(r"[/,;]", str(cell)) if p.strip()]
```

```
    return token in parts
```

```

filtered = catalog.copy()

if surec != "(Hepsi)":
    filtered = filtered[filtered["Surec"].apply(lambda x: _has_token(x, surec))]

if sinif != "(Hepsi)":
    filtered = filtered[filtered["Risk_Sinifi"].apply(lambda x: _has_token(x, sinif))]
    filtered = filtered[filtered["Risk_Sinifi"] == sinif]

if filtered.empty:
    st.warning("Bu süreç için demo veri setinde risk bulunmuyor.")

if st.session_state["ui_table"] is None or st.session_state["ui_table"].empty:
    st.session_state["ui_table"] = _init_ui_table_from_catalog(catalog)

full = st.session_state["ui_table"].copy()
ui_view = full[full["Risk_ID"].isin(filtered["Risk_ID"])].copy()
selection_locked = bool(st.session_state.get("selection_locked", False))
if selection_locked:
    ui_view = full[full["Seç"] == True].copy()

st.subheader("Risk Faktörleri Tablosu")
ui_view = ui_view.copy()
ui_view["Risk Numarası"] = ui_view["Risk_ID"]

if not selection_locked:
    show_cols = ["Risk_ID", "Risk Numarası", "Risk_Adi", "Aciklama", "Seç"]
else:

```

```
show_cols = ["Risk_ID", "Risk Numarası", "Risk_Adi", "Aciklama", "Olasilik",
"Etki", "Tespit"]
```

```
table_height = 680
```

```
with st.form("single_table_form", clear_on_submit=False):
```

```
    edited = st.data_editor(
        ui_view[show_cols],
        use_container_width=True,
        hide_index=True,
        height=table_height,
        disabled=["Risk_ID", "Risk Numarası", "Risk_Adi", "Aciklama"],
        column_config={
            "Risk_ID": None,
            "Risk    Numarası":    st.column_config.TextColumn("Risk    Numarası",
width="small"),
            "Risk_Adi": st.column_config.TextColumn("Risk Faktörü", width="medium"),
            "Aciklama": st.column_config.TextColumn("Açıklama", width="large"),
            "Seç": st.column_config.CheckboxColumn("Seç"),
            "Olasilik": st.column_config.NumberColumn(
                "Olasılık", min_value=1, max_value=10, step=1
            ),
            "Etki": st.column_config.NumberColumn("Etki", min_value=1, max_value=10,
step=1),
            "Tespit": st.column_config.NumberColumn(
                "Tespit", min_value=1, max_value=10, step=1
```

```

        ),
    },
)

c1, s1, c2, s2, c3, spacer, c4 = st.columns(
    [1.2, 0.12, 1.6, 0.12, 1.6, 6.0, 1.8],
    gap="small",
)

with c1:
    select_all = st.form_submit_button("Hepsini seç")

with c2:
    confirm = st.form_submit_button("Seçimleri onayla")

with c3:
    back = st.form_submit_button("Seçimleri düzenle")

with c4:
    run_now = st.form_submit_button("Analizi çalıştır", type="primary")

if select_all:
    ids_in_view = ui_view["Risk_ID"].tolist()
    full.loc[full["Risk_ID"].isin(ids_in_view), "Seç"] = True
    st.session_state["ui_table"] = full.sort_values("No").reset_index(drop=True)
    st.rerun()

full = st.session_state["ui_table"].copy()
if "Risk Numarası" in edited.columns:
    edited = edited.drop(columns=["Risk Numarası"])
full = _merge_edited_into_full(full, edited)

```

```
st.session_state["ui_table"] = full.sort_values("No").reset_index(drop=True)
```

```
if confirm:
```

```
    selected_ids = full.loc[full["Seç"] == True, "Risk_ID"].tolist()
```

```
    if not selected_ids:
```

```
        st.warning("Devam etmek için en az 1 risk seçmelisin.")
```

```
    else:
```

```
        st.session_state["selection_locked"] = True
```

```
        st.rerun()
```

```
if back:
```

```
    st.session_state["selection_locked"] = False
```

```
    st.rerun()
```

```
if run_now:
```

```
    if not st.session_state.get("selection_locked", False):
```

```
        st.warning("Önce seçimleri onaylamalısın.")
```

```
        return
```

```
    selected_ids = full.loc[full["Seç"] == True, "Risk_ID"].tolist()
```

```
    if not selected_ids:
```

```
        st.warning("Analiz için en az 1 risk seçmelisin.")
```

```
        return
```

```
    sel_sorted = sorted(selected_ids, key=lambda x: int(x[1:]))
```

```
    rpn_df = full[full["Risk_ID"].isin(sel_sorted)][
```

```
        ["Risk_ID", "Olasilik", "Etki", "Tespit"]
```

```
    ].copy()
```

```

with st.spinner("Analizler çalıştırılıyor..."):
    results = _compute_pipeline(sel_sorted, rpn_df)

st.session_state["results"] = results

st.session_state["page"] = "results"

st.rerun()

def page_results():
    results = st.session_state.get("results")

    if not results:
        st.warning("Önce seçim yapıp analizi çalıştırmalısın.")

    if st.button("Seçim sayfasına dön"):
        st.session_state["page"] = "input"

        st.rerun()

    return

top_left, top_right = st.columns([8, 2])

with top_left:
    st.markdown("<h3>Sonuçlar</h3>", unsafe_allow_html=True)

with top_right:
    b1, b2 = st.columns([1, 1], gap="small")

    with b1:
        go_back = st.button("Geri dön", use_container_width=True)

    with b2:
        reset = st.button("Sıfırla", use_container_width=True)

if go_back:

```

```

st.session_state["page"] = "input"

st.rerun()

if reset:

    st.session_state["results"] = None

    st.session_state["page"] = "input"

    st.session_state["selection_locked"] = False

    st.rerun()

left, right = st.columns([11, 9], gap="large")

with left:

    weighted = results.get("weighted_table")

    catalog = _load_catalog().set_index("Risk_ID")

    if weighted is None or weighted.empty:

        st.warning("Weighted tablo boş.")

    else:

        w = weighted.copy()

        w["Risk_Adi"] = w.index.to_series().map(

            lambda rid: catalog.loc[rid, "Risk_Adi"] if rid in catalog.index else ""

        )

        top5 = w.sort_values("Weighted_Driving_Power", ascending=False).head(5)

        top5_view = pd.DataFrame(

            {

                "Risk numarası": top5.index,

                "Risk Faktörü": top5["Risk_Adi"].values,

```

```

        "Final Puan": top5["Weighted_Driving_Power"].values,
        "Etki Gücü": top5["Driving_Power"].values,
        "RPN sonucu": top5["RPN"].values
        if "RPN" in top5.columns
        else [None] * len(top5),
    }
)
MAX_CH = 45
top5_view["Risk Faktörü"] = top5_view["Risk Faktörü"].astype(str).apply(
    lambda s: s if len(s) <= MAX_CH else s[: MAX_CH - 1] + "..."
)
st.dataframe(
    top5_view,
    use_container_width=True,
    hide_index=True,
    height=185,
)
micmac_path = results.get("micmac_plot_path", "")
_img_html(micmac_path, max_h_vh=50)

with right:
    ism_path = results.get("ism_plot_path", "")
    _img_html(ism_path, max_h_vh=82)

def main():

```

```
_ensure_session_defaults()

if st.session_state["page"] == "results":

    page_results()

else:

    page_input()

if __name__ == "__main__":

    main()
```